IE2106 Engineering Mathematics I

Probability & Statistics

by

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A. Introduction 5

1. Set Theory 7

B. Probability 14

1. Experiments, Sample Spaces, Events 14
2. Basic Probability 37
3. Independence of Events 55
4. Conditional Probability 70
5. Partitions, Total Probability, Bayes Theorem 91

7. Random Variables 115
8. Normal Distribution 182
9. Poisson Distribution 207
10. Binomial Distribution 230

C. Statistics 263

1. Population & Sample 264
2. Inferential Statistics 267
3. Sample Mean & Sample Variance 271

4. Parameters and Statistics **294**

5. Estimation of Parameters **296**

6. Interval Estimates - Confidence Intervals **306**

7. Confidence Interval for μ **311**

7.1. When σ is Known **315**

7.2. When σ is Unknown **339**

8. Confidence Interval for σ^2 **359**

9. Introduction to Hypothesis Testing **376**

10. Tests Concerning the Population Mean μ **401**
10.1. When σ is Known **401**
10.2. When σ is Unknown **424**

11. Tests Concerning the Population Variance σ^2 **449**

C. Statistics

- **The most essential task of statistics is to make inferences about populations on the basis of sample data.**
- **The purpose of sampling is to obtain a description of population characteristics without observing all of the population values.**
- **Sampling is also a part of our everyday consumer buying. Supermarkets frequently offer small samples of food to entice customers to try their products.**

1. Population & Sample

- A *population* consists of a collection of individual units (persons, objects, or experimental outcomes) whose characteristics are to be studied.

- A *sample* is a portion of the population that is studied to learn about the population.

Example 1.1.

(a) Quality control

A manufacturer of computer chips wants to monitor the quality of the product. Since many chips are manufactured everyday, only a sample of the chips is examined. The *population* consists of all computer chips made by the manufacturer.

(b) Pharmaceuticals

A pharmaceutical firm has developed a new drug for weight

loss. Before the firm can receive approval from the FDA to market the product, it must demonstrate the safety and effectiveness of the drug on a sample of individuals who are representative of the population for which it is intended.

(c) Polls

A US President candidate hires a polling firm to assess his chances in the upcoming election. The *population* consists of all US voters. The *sample* consists of all persons contacted by the polling firm.

2. Inferential Statistics

- Populations are usually too large to study in their entirety. It is necessary to select a representative sample of a more manageable size. This sample is then used to draw conclusions about the population.

- *Inferential statistics* consists of procedures used to make inferences (predictions, decisions) about population characteristics from information contained in a sample.

- **A type of sample that is representative of the entire population is the *random sample*.**

  A sample of n objects drawn from a population is a *random sample* if it is selected by a process in which every member of the population has essentially the same chance to be included in the sample, and that each sample of size n is equally likely to be selected.

Consider the sampling procedure:

Select every 3rd student entering this LT for math lecture.

If the population = all NTU students, then this sample is biased toward students who take math this semester, and also biased toward students who ..... :p

Summary

1. A *random sample* of n elements is a sample selected from the population in such a way that every set of n elements is as likely to be selected as any other set of n elements.

3. Sample Mean & Sample Variance

Let X be a r.v. defined on a population.

e.g. $X =$ weight of a student, population: NTU students

population mean $\mu =$ mean weight of students

population variance $\sigma^2 =$ variance of weights of students

sample mean $\bar{x} =$ mean weight of a sample of n students

sample variance $s^2 =$ variance of weights of a sample of n students

Let population size $= N$ and sample size $= n$.

- For the r.v. X, the *population mean* and *population variance* are given by

$$\mu = E(X) = \frac{1}{N}\sum_{i=1}^{N} x_i, \qquad \sigma^2 = \frac{1}{N}\sum_{i=1}^{N}(x_i - \mu)^2.$$

  σ is the *population standard deviation*.

- The *sample mean* and *sample variance* of a particular sample are given by

$$\overline{x} = \frac{1}{n}\sum_{i=1}^{n} x_i, \qquad s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \overline{x})^2.$$

s is the *sample standard deviation*.

3.1. Sample Mean

Since many different samples are possible, the sample mean is a r.v. (it can take the values of all possible sample means).

Its actual value depends on the particular sample obtained. Hence, we use $\overline{X}$ to denote the sample mean as a r.v.

- The *sampling distribution of* $\overline{X}$ is the probability distribution of $\overline{X}$.

- As a r.v., $\overline{X}$ also has a mean, $\mu_{\overline{X}}$ and a standard deviation, $\sigma_{\overline{X}}$ given by

$$\mu_{\overline{X}} = \mu, \qquad \sigma_{\overline{X}} = \frac{\sigma}{\sqrt{n}}$$

where μ is the population mean, and σ is the population standard deviation.

- If X is *normal or approximately normally distributed*, then so is $\overline{X}$. In this case, we have $\overline{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$.

- What if X is not known to be normal or approximately normally distributed?

  Then, we have *Central Limit Theorem*.

Central Limit Theorem

Suppose X is a r.v. with mean μ and standard deviation σ (> 0), defined on some non-normal population. If the sample size n is large $(n \geq 30)$ and the population size is large in comparison to n, then the sample mean $\overline{X}$ is *approximately normally distributed* with mean $\mu_{\overline{X}} = \mu$ and standard deviation $\sigma_{\overline{X}} = \sigma/\sqrt{n}$.

$$n \geq 30 \Rightarrow \overline{X} \text{ is approximately } N\left(\mu, \frac{\sigma^2}{n}\right)$$

1. sample mean $\bar{X}$ is a r.v.

$E(\bar{X}) = \mu$, $\text{Var}(\bar{X}) = \frac{\sigma^2}{n}$ $\left(\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}}\right)$

2. $X \sim N(\mu, \sigma^2)$

$\Rightarrow \bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$

for any n

3. Central Limit Theorem

$n \geq 30 \Rightarrow \bar{X}$ is approx $N\left(\mu, \frac{\sigma^2}{n}\right)$

for any X

How to ensure that $\bar{X}$ is approx normal?

* if $n < 30$, assume that X is approx normal, then $\bar{X}$ will also be approx normal.
* if $n \geq 30$, use Central Limit Thm which gives $\bar{X}$ approx normal.

** Think 3 : Refer to the sampling distribution of the sample mean.

The mean of the sample mean = population mean $= \mu$

The variance of the sample mean $= \dfrac{\text{population variance}}{n} = \dfrac{\sigma^2}{n}$

If the sample size, n, becomes very large (say, $n \to \infty$)

1. what can you conclude?

2. can you explain your conclusion, why it is reasonable? (Your explanation should not be "just let $n \to \infty$, and so ...")

Example 3.1. A study shows that the average number of pens a university student brings to class is 1.75 and the standard deviation is 0.65. In a sample of 50 university students, find the probability that the mean number of pens brought to class is more than 2.

Soln. Let $X =$ the number of pens a university student brings to class.

We are given $\mu = 1.75$ and $\sigma = 0.65$.

The sample size $n = 50\ (\geq 30)$.

By the central limit theorem, we have

$$\overline{X} \text{ is approximately } N\left(\mu, \frac{\sigma^2}{n}\right).$$

Thus, the probability required is

$$\begin{aligned} P(\overline{X} > 2) &= P\left(Z > \frac{2-\mu}{\frac{\sigma}{\sqrt{n}}}\right) \\ &= P\left(Z > \frac{2-1.75}{\frac{0.65}{\sqrt{50}}}\right) \end{aligned}$$

$$= \quad P(Z > 2.72) = 1 - P(Z \leq 2.72)$$

$$= \quad 1 - \phi(2.72) = 1 - 0.9967 = 0.0033.$$

3.2. Sample Variance

Since many different samples are possible, the sample variance is a r.v. (it can take the values of all possible sample variances). Its actual value depends on the particular sample obtained. We use S^2 to denote the sample variance as a r.v.

Theorem

Suppose random samples of size n corresponding to some r.v. X are drawn from a population whose size is much larger than n. Suppose also that X is (approximately) a normal r.v. with mean μ and standard deviation σ (> 0). Then,

$$\chi^2 \equiv \frac{(n-1)S^2}{\sigma^2}$$

is (approximately) a chi-square r.v. with $(n-1)$ degrees of freedom.

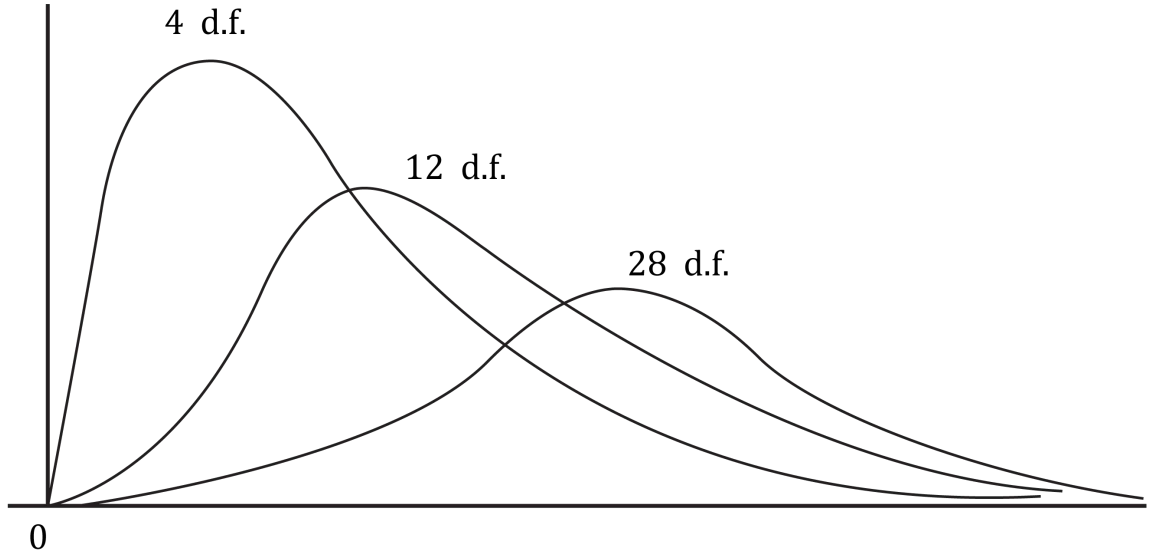
In summary,

$$X \sim N(\mu, \sigma^2)$$

$$\Rightarrow \quad \frac{(n-1)S^2}{\sigma^2} \sim \text{ chi-square with } (n-1) \text{ deg of freedom}$$

The Chi-square Distribution

The chi-square (χ^2) distribution is a continuous probability distribution. The χ^2 values begin at 0 and then are all positive. The graph of the χ^2 distribution is not symmetrical and depends on the number of degrees of freedom (d.f.). As the d.f. increase, the graph of the χ^2 distribution becomes more bell-like and begins to look more and more like the normal distribution.



Example 3.2. Suppose the times required for bus 179 to start from Interchange and arrive at EEE form a normal distribution with a standard deviation of 0.5 minute. If a sample of 17 arrival times is selected at random, find the probability that the sample variance is greater than 0.5.

Soln. Let X = time required for bus 179 to start from Interchange and arrive at EEE.

Given $\sigma = 0.5,\ n = 17$.

We know that $\chi^2 = \dfrac{(n-1)S^2}{\sigma^2}$ is a χ^2 r.v. with $(n-1) = 16$

degrees of freedom. Hence,

$$
\begin{aligned}
P(S^2 > 0.5) &= P\left(\frac{(n-1)S^2}{\sigma^2} > \frac{(n-1)(0.5)}{\sigma^2}\right) \\
&= P\left(\frac{(n-1)S^2}{\sigma^2} > \frac{(17-1)(0.5)}{(0.5)^2}\right) \\
&= P\left(\frac{(n-1)S^2}{\sigma^2} > 32\right) \\
&= P(\chi^2 > 32) = 0.01.
\end{aligned}
$$

Example *1. A certain type of electric light bulb has a mean lifetime of 1500 hours with a standard deviation of 15.5 hours.

(a) Suppose a random sample of 41 bulbs is selected for testing. What is the probability that the tested bulbs will have a mean lifetime that is at least 1505 hours?

Also, find the probability that the sample variance is greater than 174.5.

(b) Suppose a sample of 16 bulbs is selected at random, find the probability that the sample variance is less than 400.

EXTRA

Soln. (a) Let X be the lifetime of an electric light bulb.

Given that $\mu = 1500$, $\sigma = 15.5$, and the sample size is $n = 41$ (≥ 30).

Since $n > 30$, by the Central Limit Theorem, we have $\overline{X}$ is approximately normally distributed, i.e., $\overline{X}$ is approximately $N\left(\mu, \frac{\sigma^2}{n}\right)$.

EXTRA

Thus, the probability required is

$$\begin{aligned} P(\overline{X} \geq 1505) &= P\left(Z \geq \frac{1505-\mu}{\frac{\sigma}{\sqrt{n}}}\right) \\ &= P\left(Z \geq \frac{1505-1500}{\frac{15.5}{\sqrt{41}}}\right) \\ &= P\left(Z \geq 2.07\right) = 1 - P(Z < 2.07) \\ &= 1 - \phi(2.07) = 1 - 0.9808 = 0.0192. \end{aligned}$$

EXTRA

Next, assume that X is approximately normal. Then, $\chi^2 = \frac{(n-1)S^2}{\sigma^2}$ is approximately a χ^2 random variable with $(n-1) = 40$ degrees of freedom.

Thus, the probability required is

$$\begin{aligned} P(S^2 \geq 174.5) &= P\left(\frac{(n-1)S^2}{\sigma^2} \geq \frac{(n-1)(174.5)}{\sigma^2}\right) \\ &= P\left(\chi^2 \geq \frac{(40)(174.5)}{15.5^2}\right) \\ &= P\left(\chi^2 \geq 29.05\right) \\ &= 0.9. \end{aligned}$$

EXTRA

(b) Assume that X is approximately normal. Then, $\chi^2 = \frac{(n-1)S^2}{\sigma^2}$ is approximately a χ^2 random variable with $(n-1) = 15$ degrees of freedom.

$$\begin{aligned} P(S^2 < 400) &= 1 - P(S^2 \geq 400) \\ &= 1 - P\left(\frac{(n-1)S^2}{\sigma^2} \geq \frac{(n-1)(400)}{\sigma^2}\right) \\ &= 1 - P\left(\chi^2 \geq \frac{(15)(400)}{15.5^2}\right) \\ &= 1 - P\left(\chi^2 \geq 25\right) \\ &= 1 - 0.05 = 0.95. \end{aligned}$$

EXTRA

Summary

1. **Notations:**

   - μ - population mean (a fixed value)
   - σ^2 - population variance (a fixed value)
   - σ - population standard deviation (a fixed value)
   - $\overline{X}$ - sample mean, a r.v. whose actual value depends on the particular random sample obtained. Once we obtain a sample from the population we obtain $\overline{x}$ - a particular sample mean, a possible value of $\overline{X}$.

- $\mu_{\overline{X}}$ - the expected value of the r.v. $\overline{X}$ $(= \mu)$
- $\sigma^2_{\overline{X}}$ - the variance of the r.v. $\overline{X}$ $(= \sigma^2/n)$
- $\sigma_{\overline{X}}$ - the standard deviation of the r.v. $\overline{X}$ $(= \sigma/\sqrt{n})$

2. If the population is normal, $\overline{X}$ is normal regardless of the sample size.

3. If the population is not normal, $\overline{X}$ will be approximately normal if the sample size $n \geq 30$ (central limit theorem).

4. **The sample variance S^2 is a r.v. If the population is normal, then $\frac{(n-1)S^2}{\sigma^2}$ has a chi-square distribution with $(n-1)$ degrees of freedom.**

4. Parameters and Statistics

- A parameter is a numerical quantity that summarizes some feature of a population.

  E.g. the mean μ and standard deviation σ of a population r.v. X.

- A statistic is a numerical quantity that summarizes some feature of a sample.

  E.g. the mean $\overline{x}$ and standard deviation s of a random sample.

- Statistics are used to estimate parameters.

- A statistic can also be a r.v.

  A r.v. defined for random samples is called a statistic if it does not explicitly depend on any unknown population parameters.

  E.g. the sample mean $\overline{X}$ and the sample variance S^2 are statistics.

- Note that the word 'statistic' can refer to a numerical value, as in $\overline{x}$, and also to a random variable, as in $\overline{X}$. We will

know from context which meaning to attach to the word.

5. Estimation of Parameters

Suppose you are the Student Manager of NTU. The President wants to know the mean distance between home and NTU of a local NTU student ASAP in order to consider a development plan of student hostels.

You realize it is impossible to find the distance between home and NTU of every local student in the limited time allotment (ASAP). Instead you decide to be objective and

scientific by using a random sample of local students.

But what sample information will you use to estimate the mean for the entire local student population?

Possibly you said calculate the sample mean, $\overline{x}$. But how do you know some other measure would not give a closer estimate?

Another problem is explaining the accuracy and reliability of your estimate to the President. Many possible $\overline{x}$'s arise in a random sampling procedure. How close to the population mean is your estimate likely to be? Is this close enough to satisfy the President?

The following addresses questions such as these.

5.1. Types of Estimation

Estimation procedures may be divided into two types, point estimation and interval estimation.

(a) A point estimate is a single value that is calculated from a sample and is used to estimate some population parameter.

E.g. how far does the average local NTU student live from the university? The Student Manager takes a sample from the population and calculates the sample mean. Suppose a random sample of 36 students is selected and a sample mean of 5 km is calculated. This serves as the point estimate for the population mean.

The problem with point estimates is that they do not provide information about how close the estimate is to

the population parameter. The Student Manager does not know how close the point estimate of 5 km is to the actual distance the average local student lives from the university. This single value does not consider information about

- **the sample size, or**
- **the variability of the population from which the sample was selected.**

However, both of these facts will affect the accuracy of the estimate. This problem is solved through the use of interval estimates.

(b) An interval estimate of a population parameter is an interval that is expected to contain the parameter. It is a range of possible values computed from sample data.

For eg, we might want to add an 'error term' to the sample mean, leading to the statement that the estimate is $\overline{x} \pm E$, meaning that the population mean μ lies somewhere between $\overline{x} - E$ and $\overline{x} + E$. Such an estimate is an interval estimate.

5.2. Unbiased and Biased Estimators

- An estimator is a r.v. calculated from a random sample that provides either a point estimate or an interval estimate for some population parameter. E.g. the sample mean $\overline{X}$ is an estimator of the population mean μ.

- In estimating the population mean μ, obviously it is desirable to use an estimator which on average will give us a value equal to the parameter μ.

- A r.v. statistic is called an unbiased estimator of a population parameter if the expected value of the statistic is equal to the parameter. Otherwise, it is called a biased estimator.

- The sample mean $\overline{X}$ and sample variance S^2 are unbiased estimators of the corresponding population mean μ and variance σ^2, respectively, i.e.,

$$E(\overline{X}) = \mu, \qquad E(S^2) = \sigma^2.$$

Summary

1. The process of inferring the values of unknown population values (e.g., μ, σ^2) from known sample statistics (e.g., $\overline{x}$, s^2) is referred to as estimation.

2. The type of sample statistics that serves the purpose of making inferences about a given type of population parameter is called the estimator of that parameter. The estimator of a population parameter can be expressed as a single numerical value (called a point estimate) or as a range

of values within which the unknown parameter presumably lies (called an interval estimate).

3. An unbiased estimator is a sample statistic that, on the average, across many samples, takes on a value equal to the population parameter.

6. Interval Estimates – Confidence Intervals

We have seen that point estimates are useful but interval estimates convey more information. A point estimate, e.g. 5.0 for the population mean, does not indicate how much uncertainty might be associated with the estimate. On the other hand, an interval estimate provides a measure of uncertainty about the estimate and quantifies this uncertainty with a degree of confidence that the interval contains the population mean. E.g. the statement

"there is 90% confidence that the population mean is between 4.5 and 5.5"

indicates a strong likelihood of the population mean is between 4.5 and 5.5. This type of information is more useful than a simple point estimate.

A confidence interval consists of an interval estimate of a parameter accompanied by a level of confidence that the interval contains the parameter value.

What it means: The confidence is 90% ...

- **If the experiment were repeated many times, 90% of the intervals would be expected to contain μ. The remaining 10% would not be expected to contain μ.**

- **When a 90% confidence interval is computed from sample data, one can never be certain that the interval actually contains μ. Therefore, the statement is made,**

"The confidence is 90% that the interval contains μ."

Summary

1. A confidence interval is an interval estimate of a parameter, with a level of confidence that the interval contains the parameter value.

2. The idea of confidence level: for example

   "The confidence is 90% that the interval contains μ."

   This means – if the experiment were repeated many times,

   - 90% of the intervals would be expected to contain μ.

- **The remaining 10% would not be expected to contain μ.**
- **When a 90% confidence interval is computed from sample data, one can never be certain that the interval actually contains μ.**

Estimation of Population Mean μ

You: student manager

Task: find μ, the mean distance between home & NTU of local students

Deadline: asap!!

What to do: get a random sample of local students, find the sample mean $\overline{x}$, use $\overline{x}$ to estimate μ.

$\overline{x}$ is a **point estimate**. Good?

* Why use sample mean?

Because the sample mean $\overline{X}$ is an unbiased estimator of population mean μ, i.e.,

$$E(\overline{X}) = \mu$$

(Unbiased estimator: the expected value of the estimator is equal to the parameter to be estimated)

Note: sample variance S^2 is an unbiased estimator of population variance σ^2, i.e.,

$$E(S^2) = \sigma^2$$

$\overline{x}$ is a point estimate. Good?

Interval estimate

A confidence interval consists of an interval estimate of a parameter accompanied by a level of confidence that the interval contains the parameter value.

e.g. there is 90% confidence that μ lies in the interval $[4.5, 5.5]$.

What it means:

- **If the sampling process were repeated many times, 90% of the intervals obtained would be expected to contain μ. The remaining 10% would not be expected to contain μ.**

- **When a 90% confidence interval e.g. $[4.5, 5.5]$ is computed from sample data, we can never be sure that the interval actually contains μ. Therefore, the statement is made,**

"The confidence is 90% that the interval $[4.5, 5.5]$ contains μ."

7. Confidence Interval for μ

Suppose a r.v. X defined on a population has mean μ (unknown). From a random sample of size n, the value $\overline{x}$ of the sample mean $\overline{X}$ can be obtained.

7.1. When σ is Known

We require the sample mean $\overline{X}$ to be approximately normally distributed, i.e.,

$$\overline{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right) \quad \text{or} \quad Z = \frac{\overline{X}-\mu}{\sigma/\sqrt{n}} \sim N(0,1).$$

How to ensure that $\overline{X}$ is approximately normally distributed?

- **For $n < 30$, assume that X is approximately normally distributed (if this is not already given), then $\overline{X}$ will also be approximately normally distributed.**

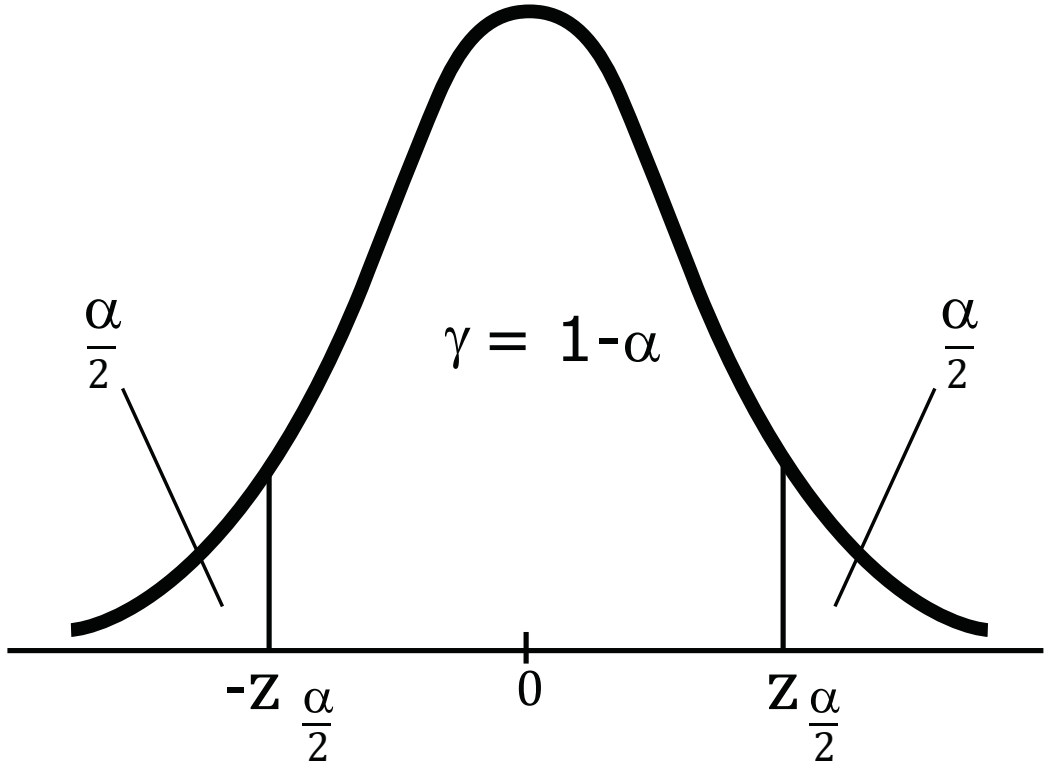
- **For $n \geq 30$, the Central Limit Theorem ensures that $\overline{X}$ is approximately normally distributed.**

Let γ be the confidence level (e.g. $\gamma = 90\%$).

Define $\alpha = 1 - \gamma$.

First, determine $z_{\alpha/2}$ from Table A1 such that

$$P\left(-z_{\alpha/2} \leq \frac{\overline{X}-\mu}{\sigma/\sqrt{n}} \leq z_{\alpha/2}\right) = \gamma.$$



Rearranging yields

$$P\left(\overline{X} - z_{\alpha/2}\,\frac{\sigma}{\sqrt{n}} \le \mu \le \overline{X} + z_{\alpha/2}\,\frac{\sigma}{\sqrt{n}}\right) = \gamma.$$

Thus, the γ confidence interval for μ is given by

$$[\overline{x} - E,\ \overline{x} + E],$$

where E is the margin of error

$$E = z_{\alpha/2} \times \frac{\sigma}{\sqrt{n}}.$$

Confidence level $= \gamma, \quad \alpha = 1 - \gamma.$

	Confidence interval for μ
σ known	• require $\overline{X}$ approximately normal • use standard normal distribution • $[\overline{x} - E,\ \overline{x} + E]$, where $E = z_{\alpha/2} \times \dfrac{\sigma}{\sqrt{n}}$

Example 7.1. A survey is carried out to gain insight of what young adults think about entrepreneurship. We are interested in the responses to the statement

" It's cool to be seen as an entrepreneur "

The possible responses are 1, 2, 3, 4, 5, with 1 = strongly disagree and 5 = strongly agree. More than 10000 responses have been collected from undergrad. The standard deviation is 0.9. A random sample of 100 responses are selected, it is found that the sample mean is 4.2. Construct a 95% confidence interval for the population mean of the response

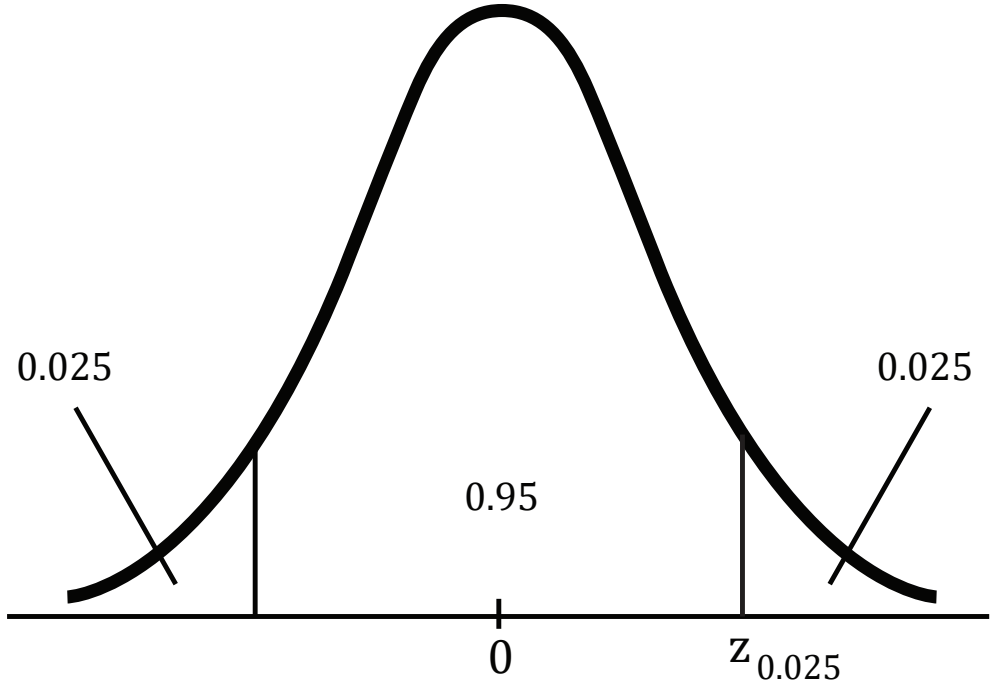
to the statement. What conclusion can you draw?

Soln. Let $X =$ score for the statement. Here,

$$\sigma = 0.9, \quad n = 100, \quad \overline{x} = 4.2, \quad \gamma = 0.95.$$

Since $n > 30$, $\overline{X}$ is approximately normally distributed by Central Limit Theorem.

1. Find the critical Z value:



From Table A1, we get $z_{0.025} = 1.96$.

2. Compute margin of error E:

$$E = z_{\alpha/2} \times \frac{\sigma}{\sqrt{n}} = 1.96 \times \frac{0.9}{\sqrt{100}} = 0.1764.$$

3. **Determine confidence interval:** The 95% confidence interval for the mean μ of X is

$$\begin{aligned}[\overline{x} - E,\ \overline{x} + E] &= [4.2 - 0.1764,\ 4.2 + 0.1764] \\ &= [4.0236,\ 4.3764].\end{aligned}$$

We conclude with 95% confidence that the population mean μ of the 10000+ scores lies somewhere between 4.02 and 4.38, i.e., the average response is Agree. There is a 5% risk that our interval will not include μ.

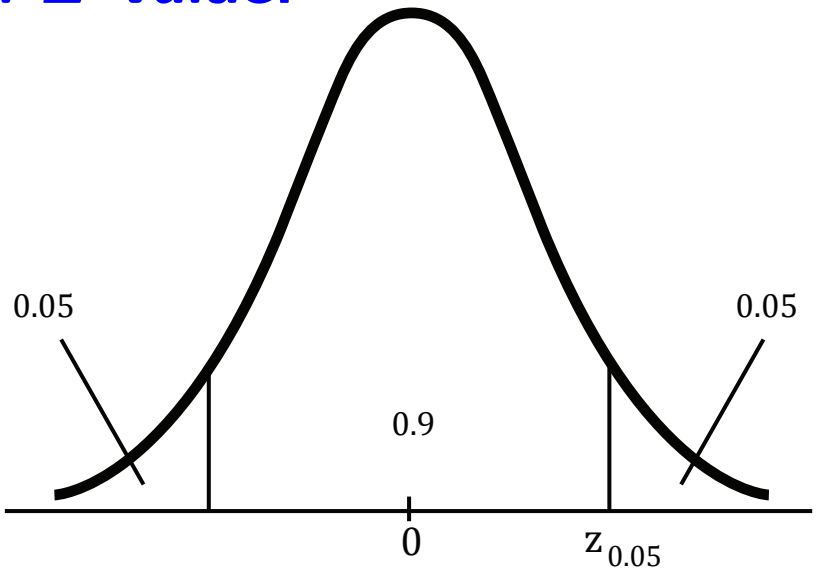
Example 7.2. **The GMAT score (Graduate Management Aptitude Test) is gaining importance in granting admission to MBA programs. GMAT scores can range from 200 to 800, they are known to have a standard deviation of 100. NUS received more than 1000 GMAT scores from MBA applicants last year. The director of admission asks that a sample of 75 scores be chosen randomly to develop a 90% confidence interval for the mean GMAT received last year. The sample mean is found to be 534. Construct the desired interval estimate.**

Soln. Let $X =$ GMAT score. Here,

$$\sigma = 100, \quad n = 75, \quad \overline{x} = 534, \quad \gamma = 0.90.$$

Since $n > 30$, $\overline{X}$ is approximately normally distributed by Central Limit Theorem.

1. Find the critical Z value:



From Table A1, we get $z_{0.05} = 1.65$.

2. **Compute margin of error E:**

$$E = z_{\alpha/2} \times \frac{\sigma}{\sqrt{n}} = 1.65 \times \frac{100}{\sqrt{75}} = 18.93.$$

3. **Determine confidence interval:** The 90% confidence interval for the mean μ of X is

$$\begin{aligned}[\overline{x} - E,\ \overline{x} + E] &= [534 - 18.93,\ 534 + 18.93] \\ &= [515.07,\ 552.93].\end{aligned}$$

The 90% confidence interval for the mean μ of X is [515.07, 552.93].

Does the population mean μ of the 1000+ scores lie somewhere between 515 and 553? It is likely, but we cannot be 100% certain. There is a 10% risk that our interval will not include μ. The confidence is 90% that our interval will include μ.

Example 7.3. **A bakery wants to estimate the mean weight of their famous chocolate chip cookies. An experiment is devised in which the weights (in g) are observed for 16 chocolate chip cookies randomly selected from the production. The weights are**

$$\begin{array}{cccccccc} 20.8 & 20.6 & 21.0 & 20.9 & 19.9 & 20.2 & 19.8 & 19.6 \\ 20.9 & 21.1 & 20.4 & 20.6 & 19.7 & 19.6 & 20.3 & 20.7 \end{array}$$

Assume the weights of cookies are adequately modelled by a normal distribution with standard deviation 0.45 g. Determine a 98% confidence interval for the average weight of the

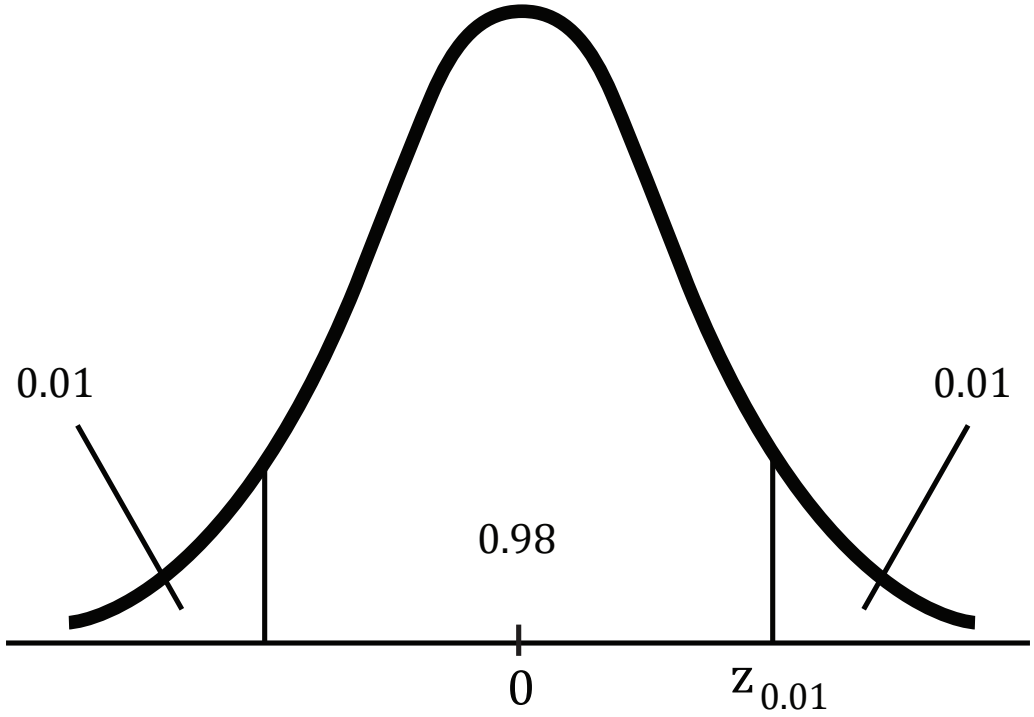
cookies.

Soln. Let $X =$ weight of a chocolate chip cookie. Here,

$$\sigma = 0.45, \quad n = 16, \quad \overline{x} = \frac{1}{16}\sum x_i = 20.38, \quad \gamma = 0.98.$$

Since X is normal, $\overline{X}$ is normally distributed.

1. Find the critical Z value:



From Table A1, we get $z_{0.01} = 2.33$.

2. Compute margin of error E:

$$E = z_{\alpha/2} \times \frac{\sigma}{\sqrt{n}} = 2.33 \times \frac{0.45}{\sqrt{16}} = 0.2621.$$

3. **Determine confidence interval:** The 98% confidence interval for the mean μ of X is

$$\begin{aligned}[\overline{x} - E,\ \overline{x} + E] &= [20.38 - 0.2621,\ 20.38 + 0.2621] \\ &= [20.12,\ 20.64].\end{aligned}$$

The bakery can be 98% confident that the cookies have an average weight between 20.12 g and 20.64 g.

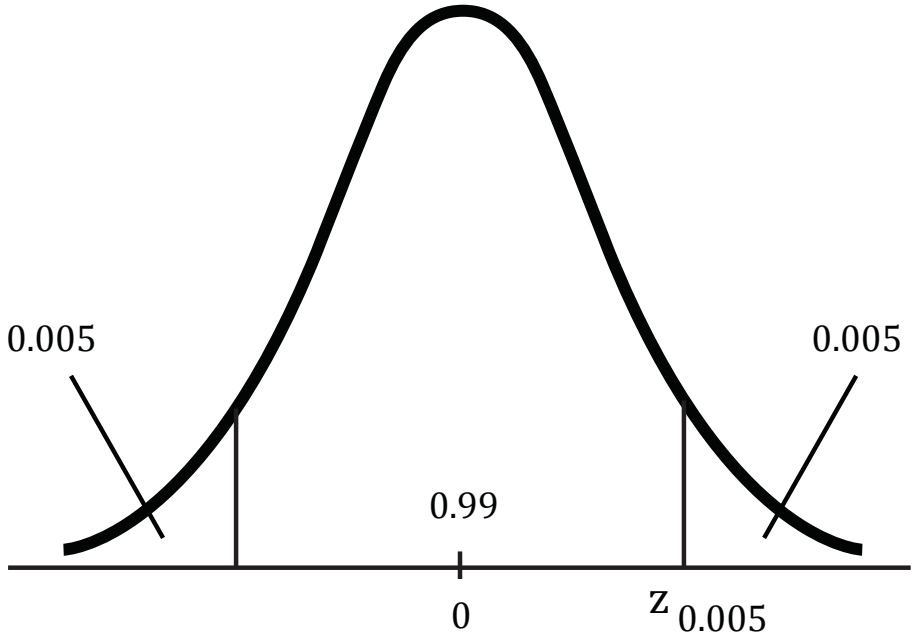
Example 7.4. An operations analyst for SIA has been asked to develop a fairly accurate estimate of the average refuelling and baggage handling time at Changi Airport. The estimate is used to arrange flight schedules so that the airline has a high percentage of "on-time" departures. The analyst randomly samples refuelling and baggage handling times of 36 flights and finds they average 24.2 minutes. The population standard deviation is 4.2 minutes. Establish a 99% confidence interval estimate.

Soln. Let $X =$ refuelling and baggage handling time at Changi

Airport. Here, $\sigma = 4.2, \quad n = 36, \quad \overline{x} = 24.2, \quad \gamma = 0.99.$

Since $n > 30$, $\overline{X}$ is approximately normally distributed by Central Limit Theorem.

1. **Find the critical Z value:**



From Table A1, we get $z_{0.005} = 2.57$.

2. **Compute margin of error E:**

$$E = z_{\alpha/2} \times \frac{\sigma}{\sqrt{n}} = 2.57 \times \frac{4.2}{\sqrt{36}} = 1.799.$$

3. **Determine confidence interval:** The 99% confidence interval for the mean μ of X is

$$[\overline{x} - E,\ \overline{x} + E] = [22.40,\ 26.00].$$

The analyst should report an interval of 22.40 min to 26.00 min at the 99% confidence level.

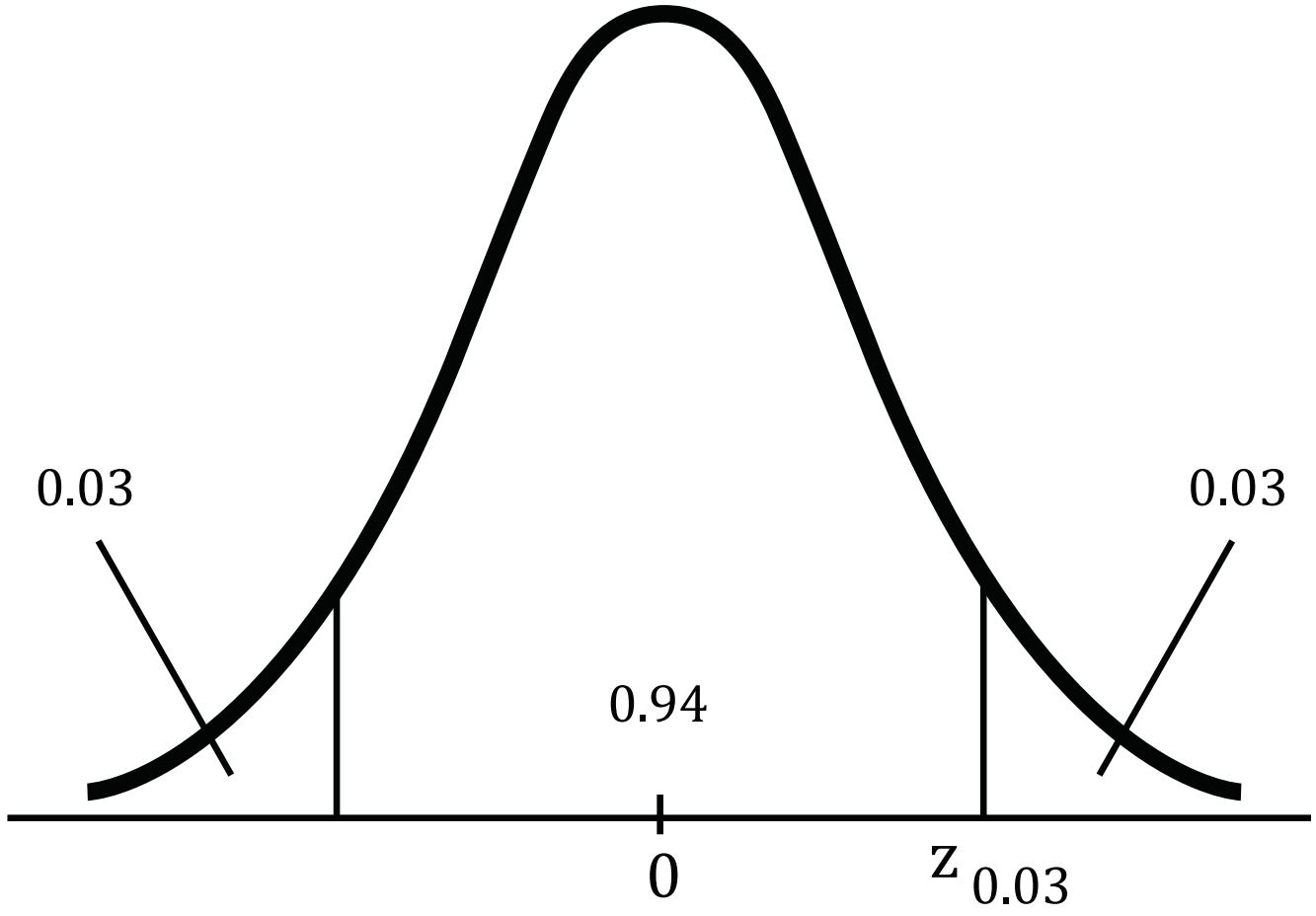
Example 7.5. The life time of a certain brand of bulb is normally distributed with standard deviation 6436 seconds. Find the sample size needed for a 94% confidence interval with a margin of error 1046 seconds.

Soln. Let $X =$ life time of a bulb. Here,

$$\sigma = 6436, \quad n = ? \ , \quad E = 1046, \quad \gamma = 0.94.$$

Since X is normal, $\overline{X}$ is normally distributed.

1. Find the critical Z value:



From Table A1, we get $z_{0.03} = 1.88$.

2. **Margin of error E:**

$$E = z_{\alpha/2} \times \frac{\sigma}{\sqrt{n}}$$

$$\Rightarrow \quad 1046 = 1.88 \times \frac{6436}{\sqrt{n}}$$

$$\Rightarrow \quad n = \left[\frac{1.88(6436)}{1046}\right]^2 = 133.8.$$

Ans: $n = 134$ (n is an integer)

Summary

Confidence level $= \gamma$, $\quad \alpha = 1 - \gamma$.

	Confidence interval for μ
σ known	• require $\overline{X}$ approximately normal • use standard normal distribution • $[\overline{x} - E,\ \overline{x} + E]$, where $E = z_{\alpha/2} \times \dfrac{\sigma}{\sqrt{n}}$

7.2. When σ is Unknown

When the standard deviation, σ, of X is not known, the sample standard deviation S is used in place of σ. Recall that

$$S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \overline{X})^2.$$

Instead of considering the statistic $Z = \frac{\overline{X}-\mu}{\sigma/\sqrt{n}}$, we now consider a new statistic

$$t = \frac{\overline{X} - \mu}{S/\sqrt{n}}.$$

What is the sampling distribution of t?

The t Distribution

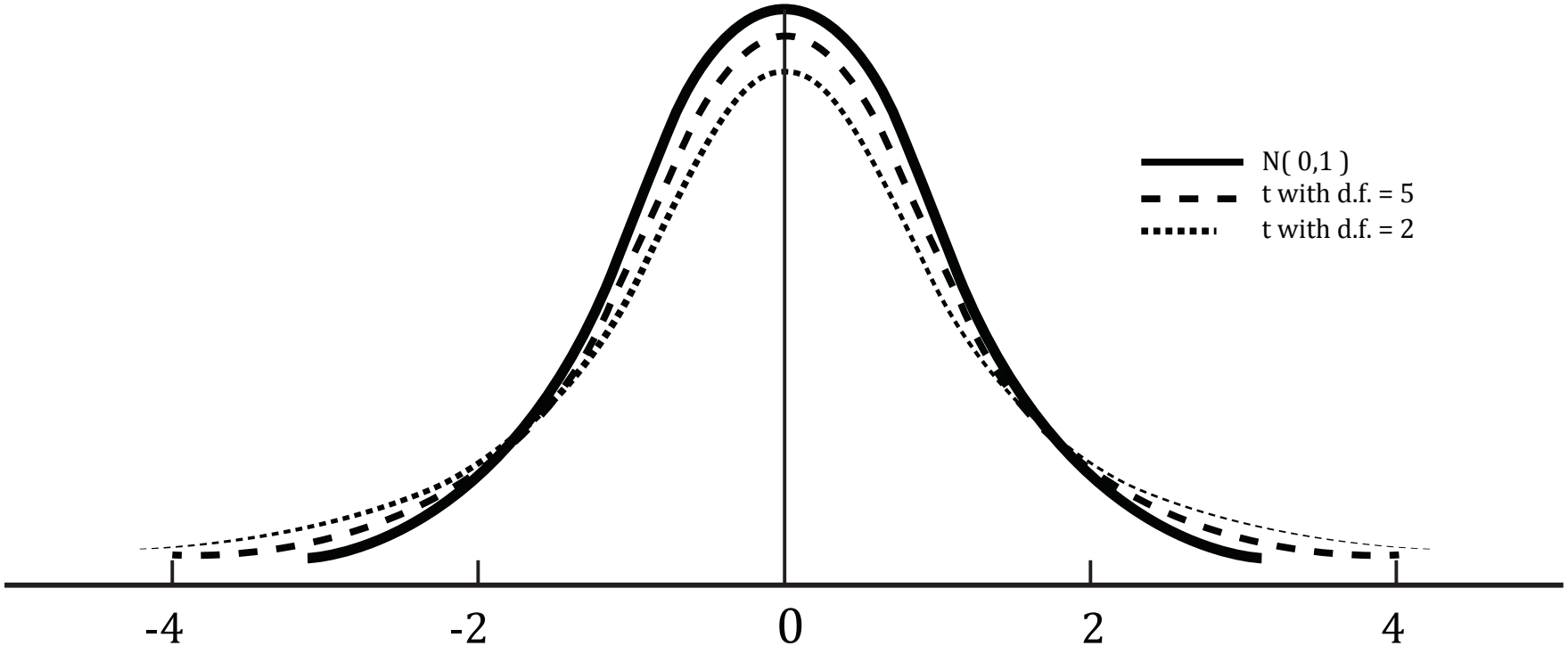
Suppose a r.v. X has mean μ. Let $\overline{X}$ be the sample mean corresponding to random samples of size n, and let S be the corresponding sample standard deviation.

If $\overline{X}$ is normally distributed, then the r.v.

$$t = \frac{\overline{X} - \mu}{S/\sqrt{n}}$$

has a t distribution with $(n-1)$ degrees of freedom.

For large n, the t distribution approaches the standard normal distribution, while for small n, the t distribution is flatter and has higher tails than the normal distribution. Note that the t distribution is symmetrical and has a mean of 0.

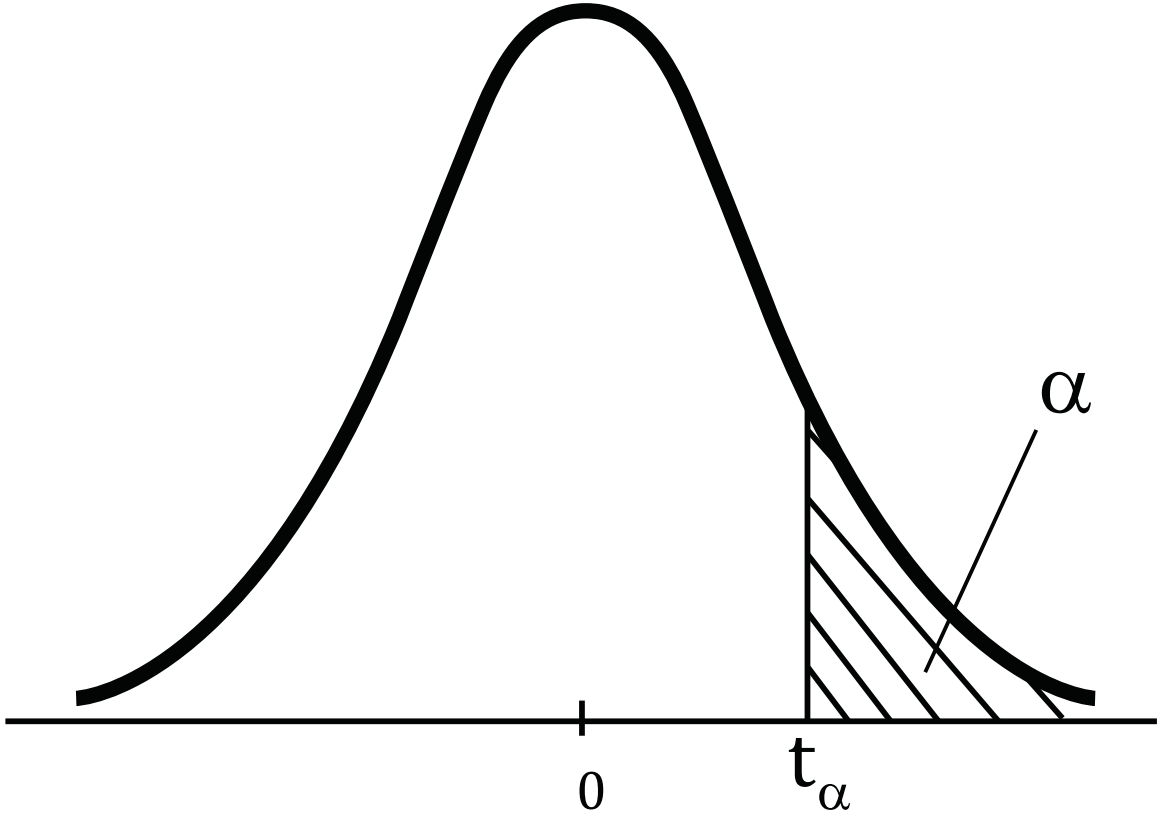


Comparison of N(0,1) and t density curves

t_α Notation

Let t_α be the no. for which the area under the t curve with $(n-1)$ d.f. to the right of t_α is α, i.e., t_α is the no. such that

$$P(t > t_\alpha) = \alpha.$$



Example 7.6. Find $t_{0.05}$, $t_{0.4}$, and '?' for d.f.=11.

Soln. From Table A3,

$$
\begin{aligned}
P(t > t_{0.05}) = 0.05 &\Rightarrow t_{0.05} = 1.796 \\
P(t > t_{0.4}) = 0.4 &\Rightarrow t_{0.4} = 0.26 \\
P(t > 0.54) = \ ? &\Rightarrow \ ? = 0.3 \ (\text{note } t_{0.3} = 0.54) \\
P(t < 0.876) = \ ? &\Rightarrow \ ? = 1 - P(t > 0.876) \\
&\Rightarrow \ ? = 1 - 0.2 = 0.8 \ (\text{note } t_{0.2} = 0.876).
\end{aligned}
$$

Example 7.7. Suppose a random sample of size 16 is selected from a normal population with $\mu = 20$. If the standard deviation of the sample is 4, find the probability $P(\overline{X} > 21.753)$.

Soln. Since the population is normal, so is $\overline{X}$. Here, σ is unknown, $\mu = 20$, $n = 16$ and $s = 4$. The statistic

$$t = \frac{\overline{X} - \mu}{S/\sqrt{n}}$$

has a t distribution with $(n-1)$ degrees of freedom.

$$
\begin{aligned}
P(\overline{X} > 21.753) &= P\left(\frac{\overline{X} - \mu}{S/\sqrt{n}} > \frac{21.753 - \mu}{S/\sqrt{n}}\right) \\
&= P\left(t > \frac{21.753 - 20}{4/\sqrt{16}}\right) \\
&= P(t > 1.753) \\
&= 0.05 \quad \text{for } d.f. = n - 1 = 15.
\end{aligned}
$$

Confidence Interval When σ is Unknown

- Here we require the sample mean $\overline{X}$ to be approximately normally distributed so that

$$t = \frac{\overline{X} - \mu}{S/\sqrt{n}} \sim t \text{ distribution with } (n-1) \text{ d.f.}$$

- For small samples ($n < 30$), this condition can be met if X is itself approximately normally distributed.

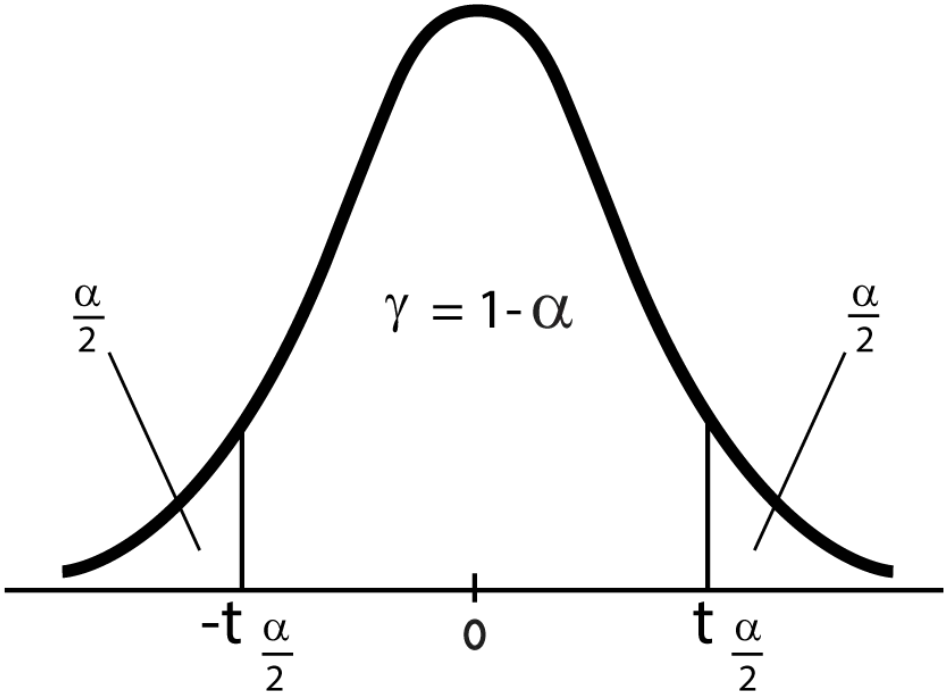
- For large samples ($n \geq 30$), the Central Limit Theorem

enables us to assume that $\overline{X}$ is approximately normally distributed regardless of the distribution of X.

Let γ be the specified confidence level. Define $\alpha = 1 - \gamma$.

First, determine $t_{\alpha/2}$ from Table A3 such that

$$P\left(-t_{\alpha/2} \leq \frac{\overline{X} - \mu}{S/\sqrt{n}} \leq t_{\alpha/2}\right) = \gamma.$$



Rearranging yields

$$P\left(\overline{X} - t_{\alpha/2}\frac{S}{\sqrt{n}} \leq \mu \leq \overline{X} + t_{\alpha/2}\frac{S}{\sqrt{n}}\right) = \gamma.$$

Thus, the γ confidence interval for μ is given by

$$[\overline{x} - E,\ \overline{x} + E],$$

where E is the margin of error

$$E = t_{\alpha/2} \times \frac{s}{\sqrt{n}}.$$

Confidence level $= \gamma$, $\quad \alpha = 1 - \gamma$.

	Confidence interval for μ
σ unknown	• require $\overline{X}$ approximately normal • use t distribution with $(n-1)$ d.f. • $[\overline{x} - E,\ \overline{x} + E]$, where $E = t_{\alpha/2} \times \dfrac{s}{\sqrt{n}}$

Confidence level $= \gamma, \quad \alpha = 1 - \gamma.$

	Confidence interval for μ
σ known	• require $\overline{X}$ approximately normal • use standard normal distribution • $[\overline{x} - E,\ \overline{x} + E]$, where $E = z_{\alpha/2} \times \frac{\sigma}{\sqrt{n}}$
σ unknown	• require $\overline{X}$ approximately normal • use t distribution with $(n-1)$ d.f. • $[\overline{x} - E,\ \overline{x} + E]$, where $E = t_{\alpha/2} \times \frac{s}{\sqrt{n}}$

Example 7.8. Our nation is concerned about the declining birthrate. 225 undergrads are randomly selected to answer the following question, on a scale of 1 (strongly disagree) to 5 (strong agree)

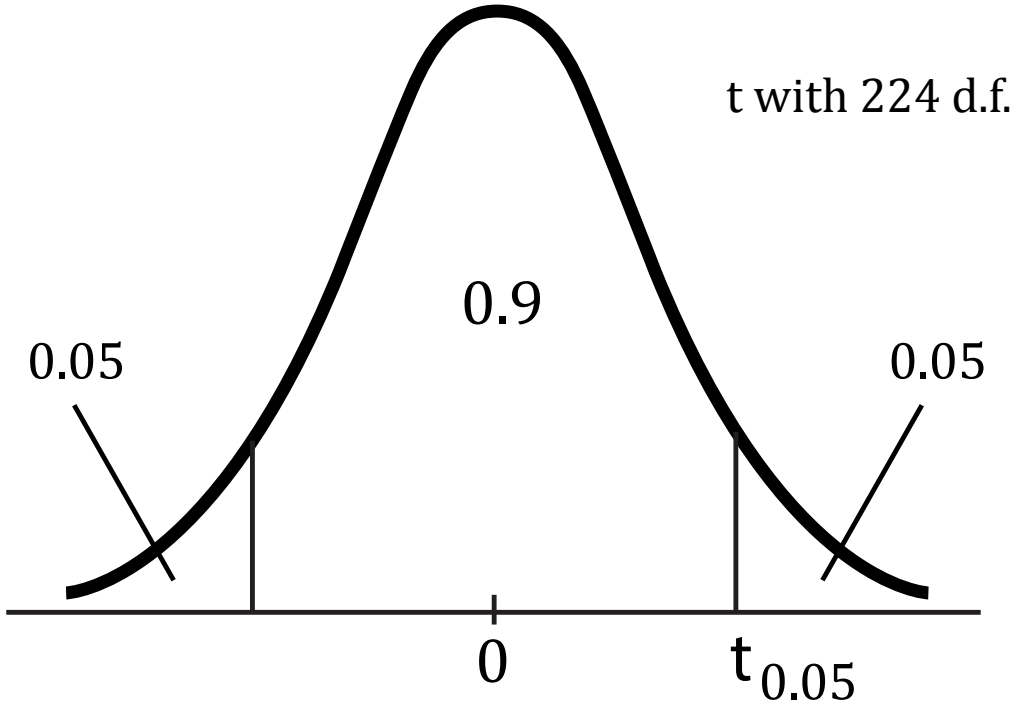
I would like to have two children, at least :p

The sample mean and standard deviation are $\overline{x} = 2.7$, $s = 1.2$. Estimate the 'mean opinion' of undergrads, using a 90% confidence level.

Soln. Let $X =$ score for the question. Here, σ is unknown, $\gamma = 0.90$, $n = 225$, $\overline{x} = 2.7$, $s = 1.2$.

Since sample size $n > 30$, by Central Limit Theorem $\overline{X}$ is approximately normally distributed.

1. Find critical t value:



Can we find $t_{\alpha/2}$ of the t r.v. with 224 d.f. ? NO. Recall that for large n, the t distribution approaches the standard normal distribution, hence we shall approximate $t_{\alpha/2}$ by $z_{\alpha/2}$.

We have $t_{0.05} \approx z_{0.05} = 1.65$.

2. **Compute margin of error E:**

$$E = t_{\alpha/2} \times \frac{s}{\sqrt{n}} = 1.65 \times \frac{1.2}{\sqrt{225}} = 0.132.$$

3. **Determine confidence interval:** The 90% confidence interval for the mean μ of X is

$$\begin{aligned}[\overline{x} - E,\ \overline{x} + E] &= [2.7 - 0.132,\ 2.7 + 0.132] \\ &= [2.568,\ 2.832].\end{aligned}$$

We conclude with 90% confidence that the 'mean opinion' of undergrads lies between 2.568 and 2.832, i.e., the average opinion is somewhere between 'neutral' and 'disagree'.

Example 7.9. An experiment has been carried out to evaluate a new sliming process. Six participants go through the whole process over a period of 3 days. The weight loss (in kg) of the participants at the end of the experiment are

$$0.46, \quad 0.61, \quad 0.52, \quad 0.48, \quad 0.57 \quad , 0.54.$$

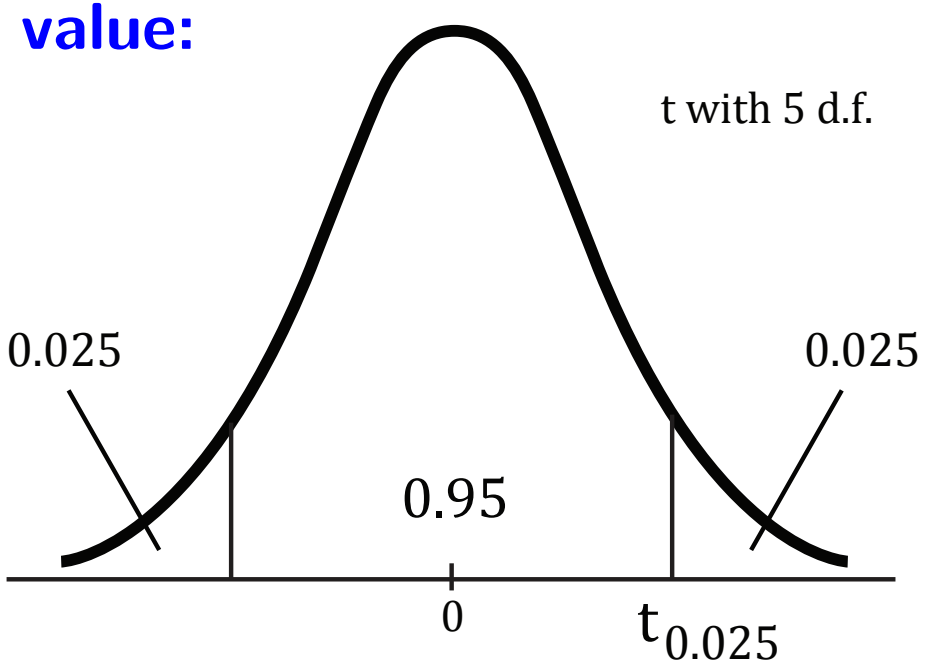
Find a 95% confidence interval for the mean weight loss resulting from the 3-day process. What is your assumption?

Soln. Let $X =$ weight loss resulting from the 3-day process.
Here, σ is unknown, $\gamma = 0.95, \ n = 6.$

Assume that X is normally distributed. Then $\overline{X}$ is normally distributed. From the sample, we calculate

$$\overline{x} = \frac{1}{6}\sum x_i = 0.53, \qquad s = \sqrt{\frac{\sum(x_i - \overline{x})^2}{n-1}} = 0.0559.$$

1. Find critical t value:



$t_{0.025} = 2.571.$

2. **Compute margin of error E:**

$$E = t_{\alpha/2} \times \frac{s}{\sqrt{n}} = 2.571 \times \frac{0.0559}{\sqrt{6}} = 0.059.$$

3. **Determine confidence interval:** The 95% confidence interval for the mean μ of X is

$$\begin{aligned}[\overline{x} - E,\ \overline{x} + E] &= [0.53 - 0.059,\ 0.53 + 0.059] \\ &= [0.471,\ 0.589].\end{aligned}$$

Summary Confidence level $= \gamma$, $\alpha = 1 - \gamma$.

	Confidence interval for μ
σ known	• require $\overline{X}$ approximately normal • use standard normal distribution • $[\overline{x} - E,\ \overline{x} + E]$, where $E = z_{\alpha/2} \times \frac{\sigma}{\sqrt{n}}$
σ unknown	• require $\overline{X}$ approximately normal • use t distribution with $(n-1)$ d.f. • $[\overline{x} - E,\ \overline{x} + E]$, where $E = t_{\alpha/2} \times \frac{s}{\sqrt{n}}$

8. Confidence Interval for σ^2

Recall the following theorem on the probability distribution of the sample variance S^2.

Theorem

Suppose random samples of size n corresponding to some r.v. X are drawn from a population whose size is much larger than n. Suppose also that X is (approximately) a normal r.v. with mean μ and standard deviation σ (> 0). Then,

$$\chi^2 \equiv \frac{(n-1)S^2}{\sigma^2}$$

is (approximately) a chi-square r.v. with $(n-1)$ degrees of freedom.

In summary,

$$X \sim N(\mu, \sigma^2)$$

$$\Rightarrow \quad \frac{(n-1)S^2}{\sigma^2} \sim \text{ chi-square with } (n-1) \text{ deg of freedom}$$

In view of the theorem, to obtain a confidence interval for σ^2, we begin with a r.v. X that is approximately normally distributed with mean μ and unknown variance σ^2.

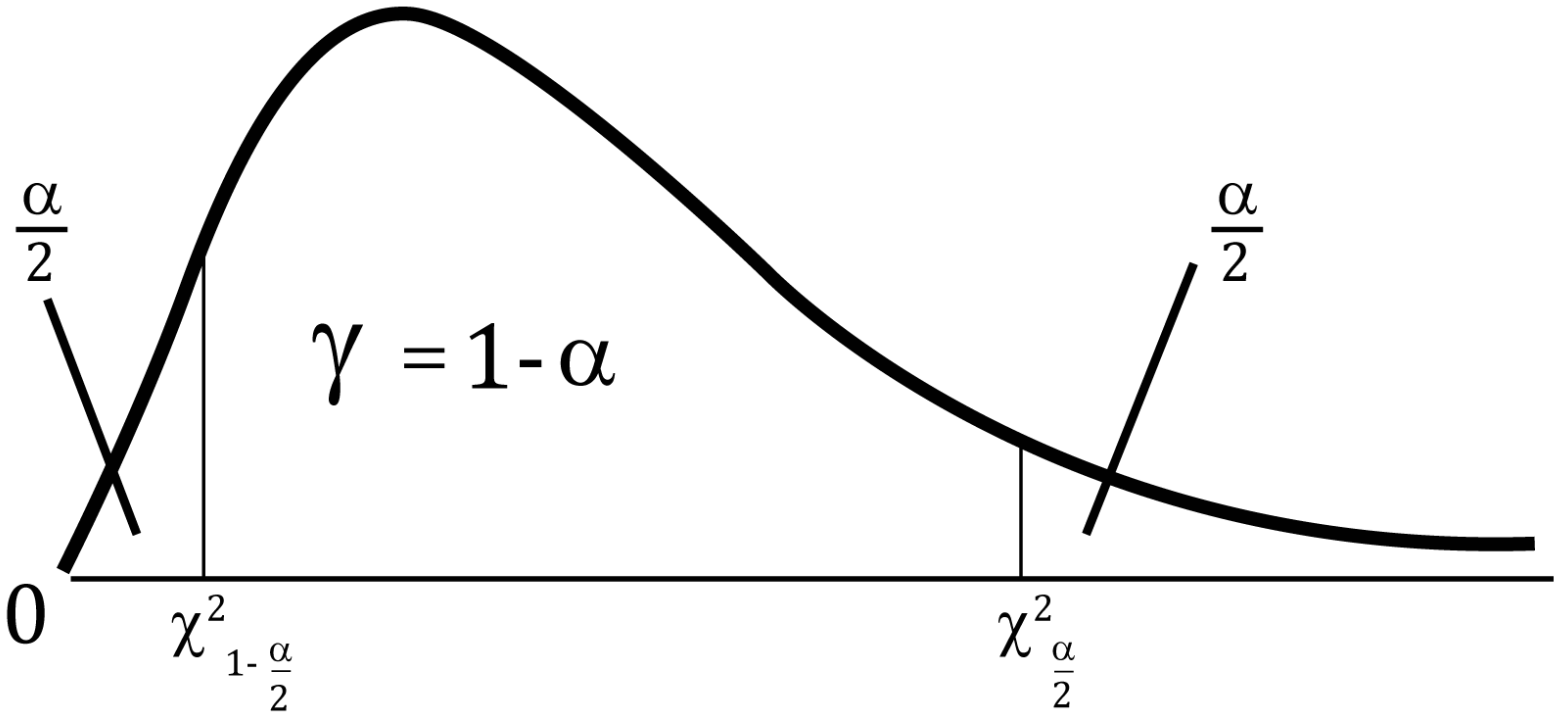
The confidence intervals depend on the chi-square distribution, which is not symmetric, so they do not take the usual form $s^2 \pm E$, where s^2 is a value of the sample variance S^2 obtained in a random sample.

Let γ be the confidence level.

Define $\alpha = 1 - \gamma$.

Consider χ^2 r.v. with $(n-1)$ d.f. We have

$$P(\chi^2_{1-\alpha/2} \leq \chi^2 \leq \chi^2_{\alpha/2}) = \gamma.$$



Note: $P(\chi^2 \geq \chi^2_{\alpha/2}) = \frac{\alpha}{2}$, $P(\chi^2 \geq \chi^2_{1-\alpha/2}) = 1 - \frac{\alpha}{2}$.

Replacing χ^2 by $\frac{(n-1)S^2}{\sigma^2}$, we get

$$P\left(\chi^2_{1-\alpha/2} \leq \frac{(n-1)S^2}{\sigma^2} \leq \chi^2_{\alpha/2}\right) = \gamma$$

which is equivalent to

$$P\left(\frac{(n-1)S^2}{\chi^2_{\alpha/2}} \leq \sigma^2 \leq \frac{(n-1)S^2}{\chi^2_{1-\alpha/2}}\right) = \gamma.$$

Therefore, a γ confidence interval for σ^2 is

$$\left[\frac{(n-1)S^2}{\chi^2_{\alpha/2}}, \frac{(n-1)S^2}{\chi^2_{1-\alpha/2}}\right]$$

Taking square roots gives a γ confidence interval for σ

$$\left[\sqrt{\frac{(n-1)S^2}{\chi^2_{\alpha/2}}}, \sqrt{\frac{(n-1)S^2}{\chi^2_{1-\alpha/2}}}\right]$$

Confidence level $= \gamma, \quad \alpha = 1 - \gamma.$

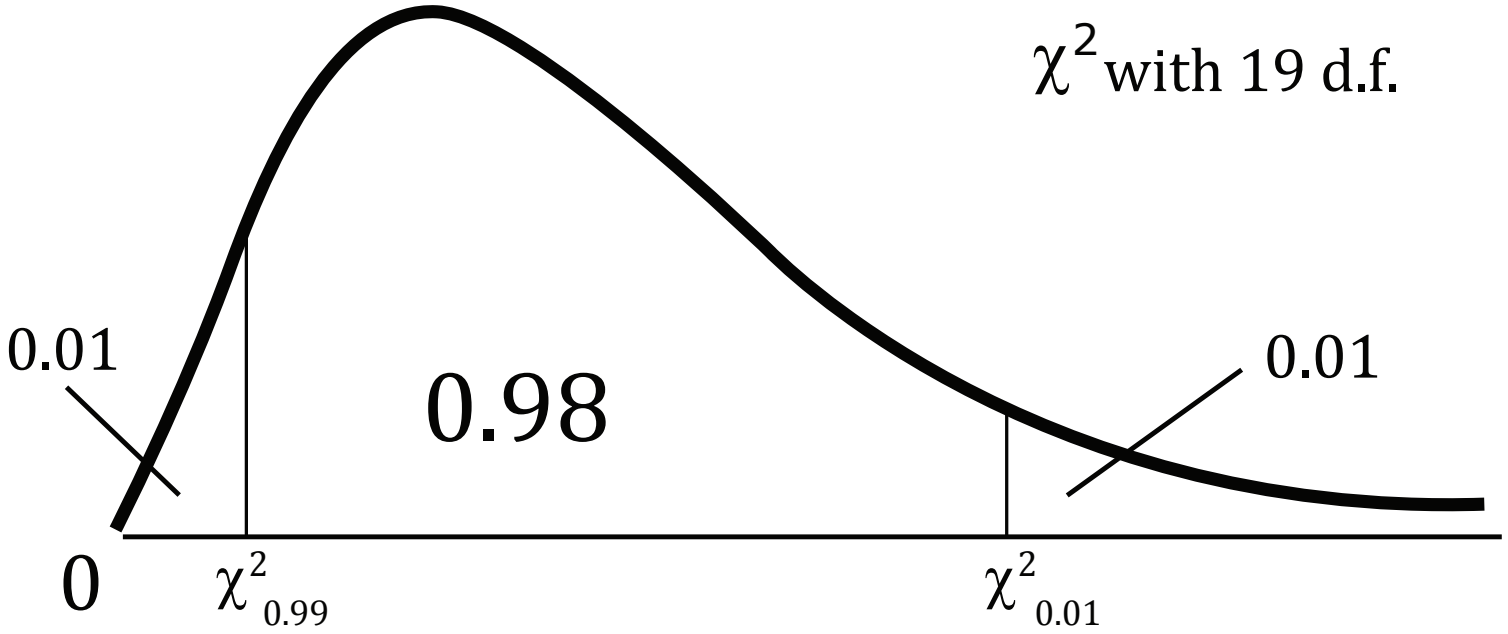
	Confidence interval for σ^2
	• **require** X **approximately normal** • **use** χ^2 **distribution with** $(n-1)$ **d.f.** • $\left[\dfrac{(n-1)s^2}{\chi^2_{\alpha/2}}, \dfrac{(n-1)s^2}{\chi^2_{1-\alpha/2}}\right]$

Example 8.1. The variation of the potency of a drug must be small for safety reasons. A pharmaceutical company was interested in determining the variance of potency measurements for a certain painkiller. Towards this end, a random sample of 20 painkillers produced a variance equal to 0.0018. Construct a 98% confidence interval for the variance of the potency measurements of painkillers.

Soln. Let $X =$ potency measurement of painkiller.

Assume that X is normally distributed.

1. $\gamma = 0.98$. **Find the critical values from** χ^2 **with d.f.** $= n - 1 = 20 - 1 = 19$.



$$\chi^2_{\alpha/2} = \chi^2_{0.01} = 36.19, \quad \chi^2_{1-\alpha/2} = \chi^2_{0.99} = 7.63.$$

2. **The 98% confidence interval for the variance of the potency measurements of painkillers is**

$$\left[\frac{(n-1)s^2}{\chi^2_{\alpha/2}},\ \frac{(n-1)s^2}{\chi^2_{1-\alpha/2}}\right] = \left[\frac{19(0.0018)}{36.19},\ \frac{19(0.0018)}{7.63}\right]$$

$$= [0.000945,\ 0.00448].$$

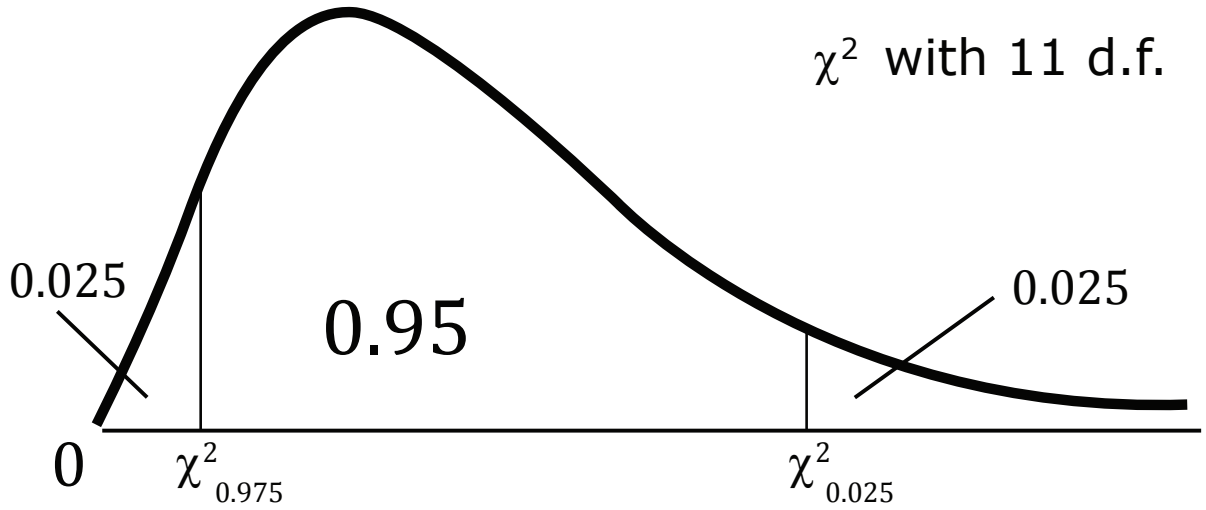
Example 8.2. Twelve cheese burgers are selected from a McDonald's outlet. The cheese burgers are weighed, and the results are as follows (in oz):

$$32.6,\ 33.1,\ 32.1,\ 32.7,\ 31.9,\ 31.8$$
$$32.4,\ 32.3,\ 32.8,\ 33.3,\ 32.7,\ 32.6$$

Assume that the weights of the cheese burgers are normally distributed. Find a 95% confidence interval for σ.

Soln. Let $X =$ weight of a cheese burger. X is normally distributed (given).

1. $\gamma = 0.95$. Find the critical values from χ^2 with d.f. $= n - 1 = 12 - 1 = 11$.



$$\chi^2_{\alpha/2} = \chi^2_{0.025} = 21.92, \quad \chi^2_{1-\alpha/2} = \chi^2_{0.975} = 3.81.$$

2. From the given data, the sample variance is

$$s^2 = \frac{\sum (x_i - \overline{x})^2}{n-1} = 0.2039.$$

3. The 95% confidence interval for the variance σ^2 of the weights of cheese burgers is

$$\left[\frac{(n-1)s^2}{\chi^2_{\alpha/2}}, \frac{(n-1)s^2}{\chi^2_{1-\alpha/2}}\right] = \left[\frac{11(0.2039)}{21.92}, \frac{11(0.2039)}{3.81}\right]$$

$$= [0.1023,\ 0.589].$$

The 95% confidence interval for the standard deviation σ of the weights of cheese burgers is

$$[\sqrt{0.1023},\ \sqrt{0.589}] = [0.32,\ 0.767].$$

Summary Confidence level $= \gamma$, $\alpha = 1 - \gamma$.

	Confidence interval for σ^2
	• require X approximately normal • use χ^2 distribution with $(n-1)$ d.f. • $\left[\dfrac{(n-1)s^2}{\chi^2_{\alpha/2}}, \dfrac{(n-1)s^2}{\chi^2_{1-\alpha/2}}\right]$

Small Sample and X is Not Normal

All confidence intervals obtained previously (for $n < 30$) are based on the requirement that X is normal. So what happens when X is not quite normal?

Studies have shown that even when X is not quite normal, the distribution of the r.v. $\frac{\overline{X}-\mu}{S/\sqrt{n}}$ will have a shape much like that of a t distribution with $(n-1)$ d.f. Hence, the confidence intervals for mean using the t distribution are very close to the true confidence intervals when X is not quite normal.

However, confidence intervals obtained for the variance can deviate very significantly from the true confidence intervals when X is not normally distributed. If X is not normal, the r.v. $\frac{(n-1)S^2}{\sigma^2}$ may have a sampling distribution quite different from the chi-square distribution with $(n-1)$ d.f.

9. Introduction to Hypothesis Testing

Hypothesis testing is a method of analyzing data to choose between 2 competing statements or hypotheses. In the previous section, confidence intervals were prescribed for means and variances. Here we discuss another type of statistical inference regarding these same parameters.

Example

Suppose we are interested to know the average IQ of all students of NTU. We may assume, or hypothesize, that the

average IQ of all NTU students is $\mu = 115$.

How to know whether our hypothesis is true or not? Take sample. The purpose of taking a sample is to draw some conclusion about the population. Upon collecting the sample data, we want to know what they are telling us about the population.

Suppose we take a sample of NTU students and find that the sample mean of their IQ is $\overline{x} = 116$. Could we conclude that our hypothesis of $\mu = 115$ is wrong? After all, $115 \neq 116$! Of course, we could not. Although 116 is not

115, it is very close. It would be quite possible to select a sample with a mean of $\overline{x} = 116$ from a population with a mean of $\mu = 115$. This small difference could be attributed to chance. We might just happen to choose for our sample a few of the students who were a little more intelligent on the average than the rest. This would have the effect of slightly inflating the sample mean above the population mean. It would in no way suggest that the population mean was not 115.

On the other hand, suppose our sample mean turned out

to be $\overline{x} = 150$, it would be difficult to explain that this large difference is merely attributed to chance. We might have to conclude that there is 'significant difference' between $\overline{x} = 150$ and what we thought the population mean was ($\mu = 115$). 'Significant difference' - a difference which cannot be explained by sampling error (the natural variation among samples taken from the same population). A result as high as $\overline{x} = 150$ might suggest that the reason we observed such a high sample mean is because the population from which the sample was taken has a mean more than our hypothesized value of 115.

The above illustration is a decision question about a population characteristic (mean). This question can be structured as a test of hypotheses.

What is hypothesis testing?

* **Hypothesis testing is a process of using sample data and statistical procedure to decide whether to reject or not to reject a hypothesis (statement) about a population parameter value.**

* **A statistical hypothesis is a statement to the effect that**

the population possesses some particular property. The symbol H is used to denote a statistical hypothesis. Just as every event A determines a complementary event A', called 'not A', so any hypothesis H determines a complementary hypothesis which says that the population does NOT possess the property asserted by H. Thus any hypothesis testing problem involves using the sample data to choose between 2 complementary hypotheses. Traditionally, one of these is called the null hypothesis (H_0) and the other is called the alternative hypothesis (H_1).

* The null hypothesis (H_0) is generally an assumption about the population which we do not wish to reject unless there is a convincing or 'significant' amount of evidence against it.

The alternative hypothesis (H_1) is just the contrary of H_0.

* It is sometimes difficult to decide what hypothesis should be H_0 and what should be the alternative H_1. The choice depends on what question we are trying to answer. Essentially, the hypothesis test answers the question:

Is there convincing evidence against H_0?

i.e., is there convincing evidence in favor of H_1?

So, in order to formulate H_0 and H_1, we should try to write the question of interest in the form:

Is there convincing evidence that H is true?

and then take H to be the alternative hypothesis H_1.

Note that this is NOT equivalent to asking if there is convincing evidence against H_1, as there may be no convincing evidence either way. That is why the choice of H_0 does make a difference to the conclusions.

9.1. Hypothesis Testing

Hypothesis testing is a process of using sample data and statistical procedure to choose between 2 competing statements or hypotheses.

Null hypothesis vs **Alternative hypothesis**

H_0 vs H_1

H_0 and H_1 are complementary to each other.

H_0 : following most books, always write $=$, e.g. $\mu = 15$

H_1 : 3 choices $<$, $>$, $\neq$, e.g. $\mu \neq 15$, $\mu > 15$, $\mu < 15$

- It is easy to write H_0, since it is always written as $=$, even if we do not mean it.
- How to decide on the correct H_1?

  We should ask ourself the question:

  Is there convincing evidence that H is true?

  Take H to be the alternative hypothesis H_1.

9.2. Steps of Hypothesis Testing – Example

Toyota Corolla Altis - excellent fuel consumption!

Marketing manager of Toyota claims

Corolla Altis will average over 15 km per litre of petrol.

We want to investigate the claim.

Step 1. Formulate the hypotheses

Let $X =$ the fuel consumption of Corolla Altis, i.e., the number of km run per litre of petrol. Let μ be the mean of X.

$H_0 : \mu = 15$

$H_1 : \mu > 15$

Step 2. **Determine the rejection region (critical region) for H_0**

Is there a significant difference between $\overline{x}$ and H_0?

* **A significant difference strongly suggests that H_0 is not true.**

* We introduce level of significance, α for the test.

The level of significance α represents the extent to which H_0 is to be protected. Specifically, the probability of rejecting H_0 when it is true must not exceed α, i.e.,

$$P(\text{reject } H_0 \mid H_0 \text{ is true}) \leq \alpha.$$

Obviously, α should be small, e.g. $\alpha = 0.05(5\%)$ or 0.01 (1%).

* **We can now determine the rejection region for H_0. Depending on the type of test, we have 3 different situations.**

- **Upper-tail test** $H_0: \mu \leq \mu_0, \qquad H_1: \mu > \mu_0$
- **Lower-tail test** $H_0: \mu \geq \mu_0, \qquad H_1: \mu < \mu_0$
- **Two-tail test** $H_0: \mu = \mu_0, \qquad H_1: \mu \neq \mu_0$

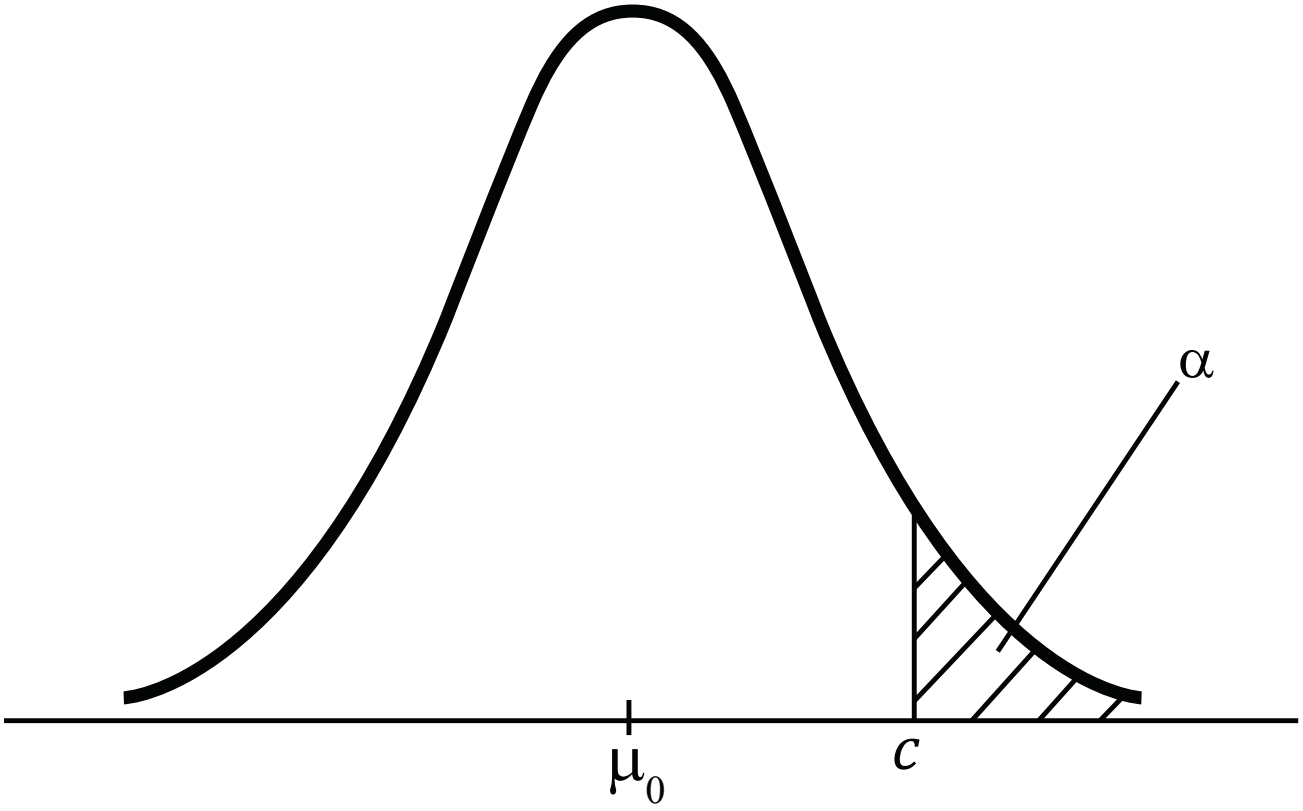
- **Upper-tail test** $H_0: \mu \leq \mu_0, \qquad H_1: \mu > \mu_0$

The rejection region is given by the set of outcomes

$$\{\overline{X} > c\}$$

where the value c, called the critical value, is defined by

$$P(\overline{X} > c) = \alpha.$$



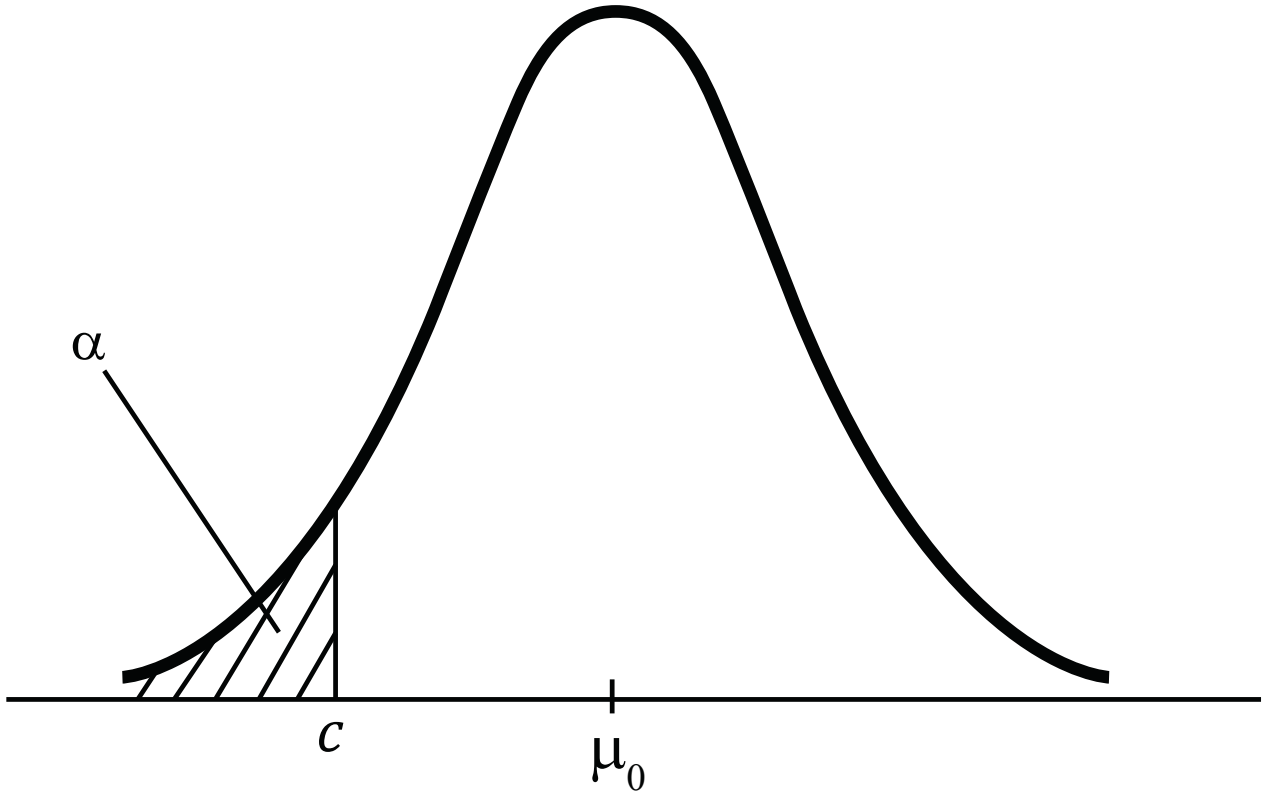
- **Lower-tail test** $H_0: \ \mu \geq \mu_0, \qquad H_1: \ \mu < \mu_0$

  The rejection region is given by the set of outcomes

$$\{\overline{X} < c\}$$

  where the critical value c is defined by

$$P(\overline{X} < c) = \alpha.$$



- **Two-tail test** $H_0: \mu = \mu_0, \qquad H_1: \mu \neq \mu_0$

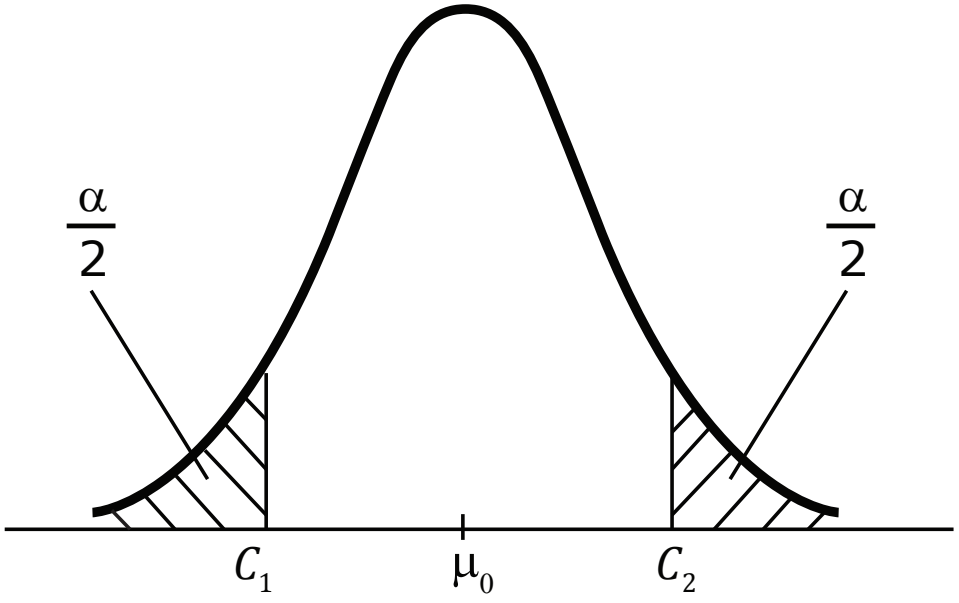
The rejection region is given by the set of outcomes

$$\{\overline{X} < c_1\} \cup \{\overline{X} > c_2\}$$

where the critical values c_1, c_2 are defined by

$$P(\overline{X} < c_1) = P(\overline{X} > c_2) = \frac{\alpha}{2}.$$

In the case of a symmetrical distribution, e.g. normal or t distribution, $c_1 = -c_2$.



* Returning to our example, we have

$$H_0: \quad \mu = 15 \qquad H_1: \quad \mu > 15.$$

Suppose that $\alpha = 0.05$, i.e., we are testing at the **5% level of significance**.

This is an upper-tail test with the critical value c defined by

$$P(\overline{X} > c) = \alpha.$$

or

$$P(Z > c) = 0.05.$$

From Table A1, we find that the critical value is $c = 1.65$.

Step 3. Collect data, calculate the test statistic

* Suppose a random sample of 49 Corolla Altis gives an average $\overline{x} = 16$ km/litre. Suppose the population standard deviation is $\sigma = 2.9$ km/litre.

Suppose the true mean were $\mu_0 = 15$ km/litre. Then, by the Central Limit Theorem

$$\overline{X} \sim N\left(\mu_0, \frac{\sigma^2}{n}\right) = N\left(15, \frac{(2.9)^2}{49}\right).$$

The test statistic for testing hypothesis about μ, with σ known, is

$$Z = \frac{\overline{X} - \mu_0}{\sigma/\sqrt{n}}.$$

The value of the test statistic calculated from the sample is

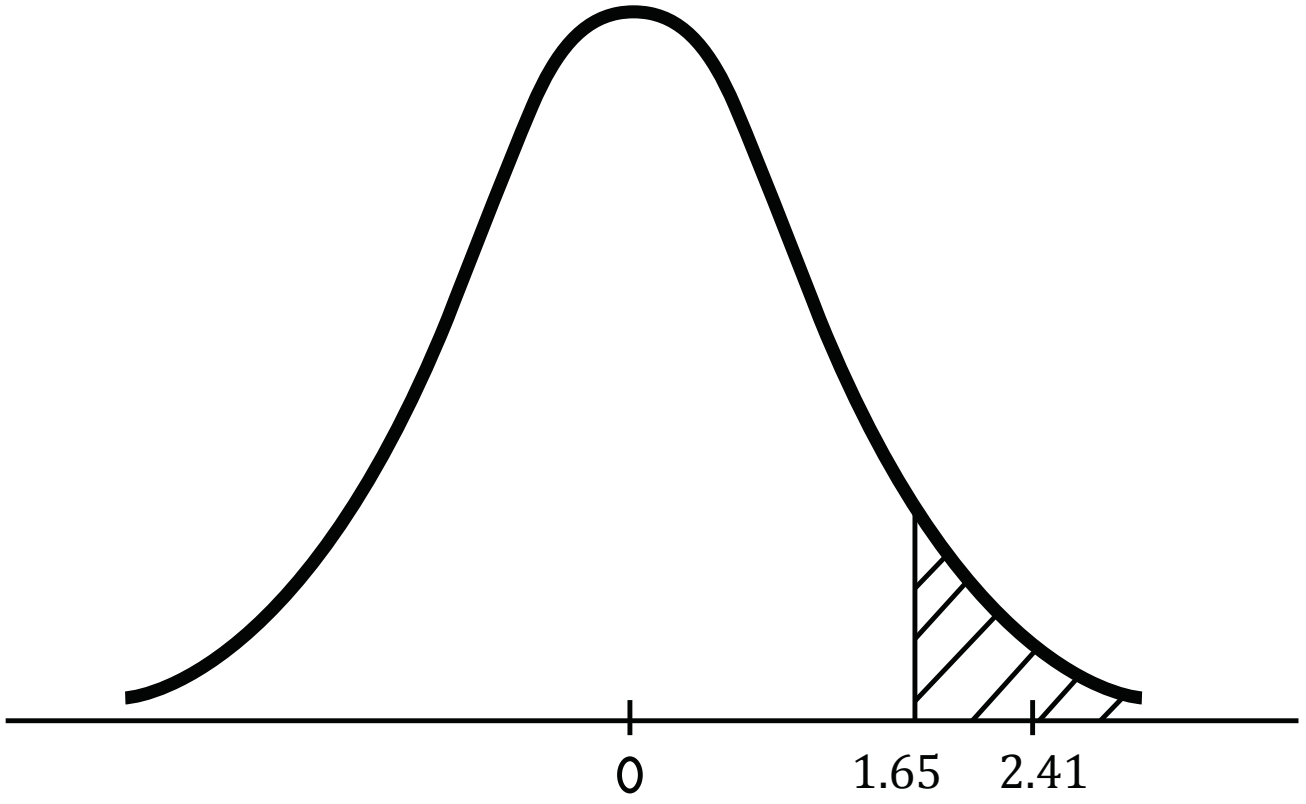
$$z = \frac{\overline{x} - \mu_0}{\sigma/\sqrt{n}} = \frac{16 - 15}{2.9/\sqrt{49}} = 2.41.$$

Step 4. Accept or reject H_0, draw conclusions

* If the value of the test statistic falls within the critical region (rejection region), then we say that, at significant level α, there is significant evidence against H_0, we reject H_0 (and accept H_1).

* If the value of the test statistic does not fall within the critical region, then we say that, at significant level α, there is no significant evidence against H_0, we accept H_0 (and reject H_1).

* **In our example, the value of the test statistic is $z = 2.41$ which falls within the critical region defined by $z > 1.65$. Hence, at significant level $\alpha = 0.05$, we will reject H_0 (and accept H_1) - we conclude that Corolla Altis average above 15 km/litre.**



Summary

1. H_0 vs H_1 (complementary)

   H_0 : always write $=$

   H_1 : 3 choices $<$, $>$, $\neq$

2. H_1 is the statement that you want to test.

3. Four steps of hypothesis testing

(I) Formulate H_0 and H_1

(II) Calculate test statistic

(III) Draw rejection region (critical region)

(IV) Conclusion: reject ? accept ?

Write in words about H_1

Steps (II) & (III) are interchangeable.

4. Three types of test

Upper-tail test $H_1 : \mu > \mu_0$

Lower-tail test $H_1 : \mu < \mu_0$

Two-tail test $H_1 : \mu \neq \mu_0$

10. Tests Concerning the Population Mean μ

$$H_0: \quad \mu = \mu_0$$

10.1. When σ is Known

Let X be a r.v. with unknown mean μ and known standard deviation σ. We require

$\overline{X}$ to be approximately normally distributed.

Thus,

$$\overline{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right).$$

The appropriate test statistic is the standardized sample mean, namely,

$$Z = \frac{\overline{X} - \mu_0}{\sigma/\sqrt{n}}$$

which, assuming H_0 is true, is the standard normal r.v.

Note How to ensure that $\overline{X}$ is approximately normally distributed?

- If we do NOT know that X is normally distributed, then provided that $n \geq 30$, the Central Limit Theorem tells us that $\overline{X}$ is approximately normally distributed.

- If $n < 30$, then we must assume that X is normal so that $\overline{X}$ is also normal.

Recall the steps of hypothesis testing:

(0) Make sure that the requirement is satisfied.

"$\overline{X}$ **is approximately normal**"

(I) Formulate H_0 and H_1

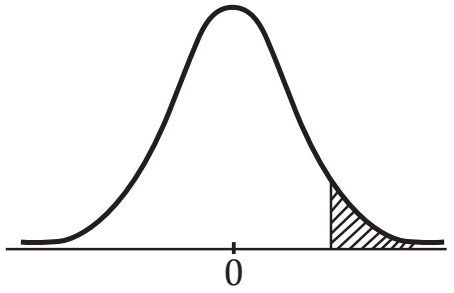
$$H_0 : \mu = \mu_0 \qquad H_1 : \mu \qquad \mu_0$$

(II) Calculate test statistic

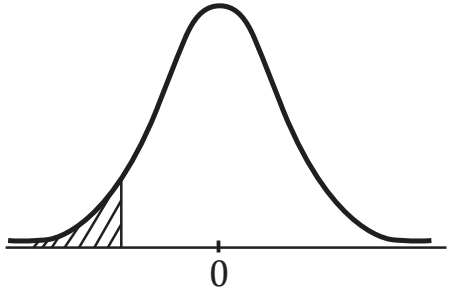
$$Z = \frac{\overline{X} - \mu_0}{\sigma / \sqrt{n}}$$

(III) Draw rejection region (critical region)

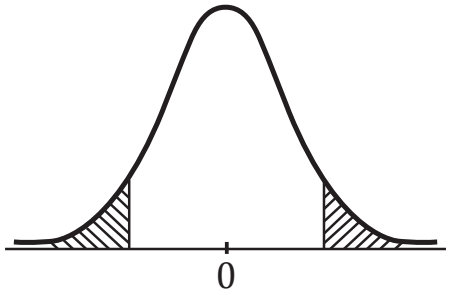
Upper-tail test $H_1 : \mu > \mu_0$



Lower-tail test $H_1 : \mu < \mu_0$



Two-tail test $H_1 : \mu \neq \mu_0$



(IV) Conclusion: reject ? accept ? Write in words about H_1

	Hypothesis testing for μ $\quad H_0 : \mu = \mu_0$
σ known	• require $\overline{X}$ approximately normal • test statistic $Z = \dfrac{\overline{X} - \mu_0}{\sigma/\sqrt{n}}$

Example 10.1. An engineer at a car service centre has designed a new procedure that he thinks will shorten a standard 5-minute checking time. In an experiment with the new procedure, a random sample of 36 cars required an average of 4.85 minutes for checking. The standard deviation of checking times is known to be 0.5 minutes. Test whether the new procedure has reduced the mean checking time to less than 5 minutes using a 1.5% level of significance.

Soln. Let $X =$ checking time (in min). X has a standard deviation of $\sigma = 0.5$ min. $\alpha = 0.015$.

Since $n = 36 > 30$, by CLT $\overline{X}$ is approximately normal.

1. The engineer is testing to see whether the data display a reduced time. Thus, the alternative hypothesis is stated as 'less than'.

   $H_0 : \ \mu = 5, \qquad H_1 : \ \mu < 5$ **(lower-tail test)**

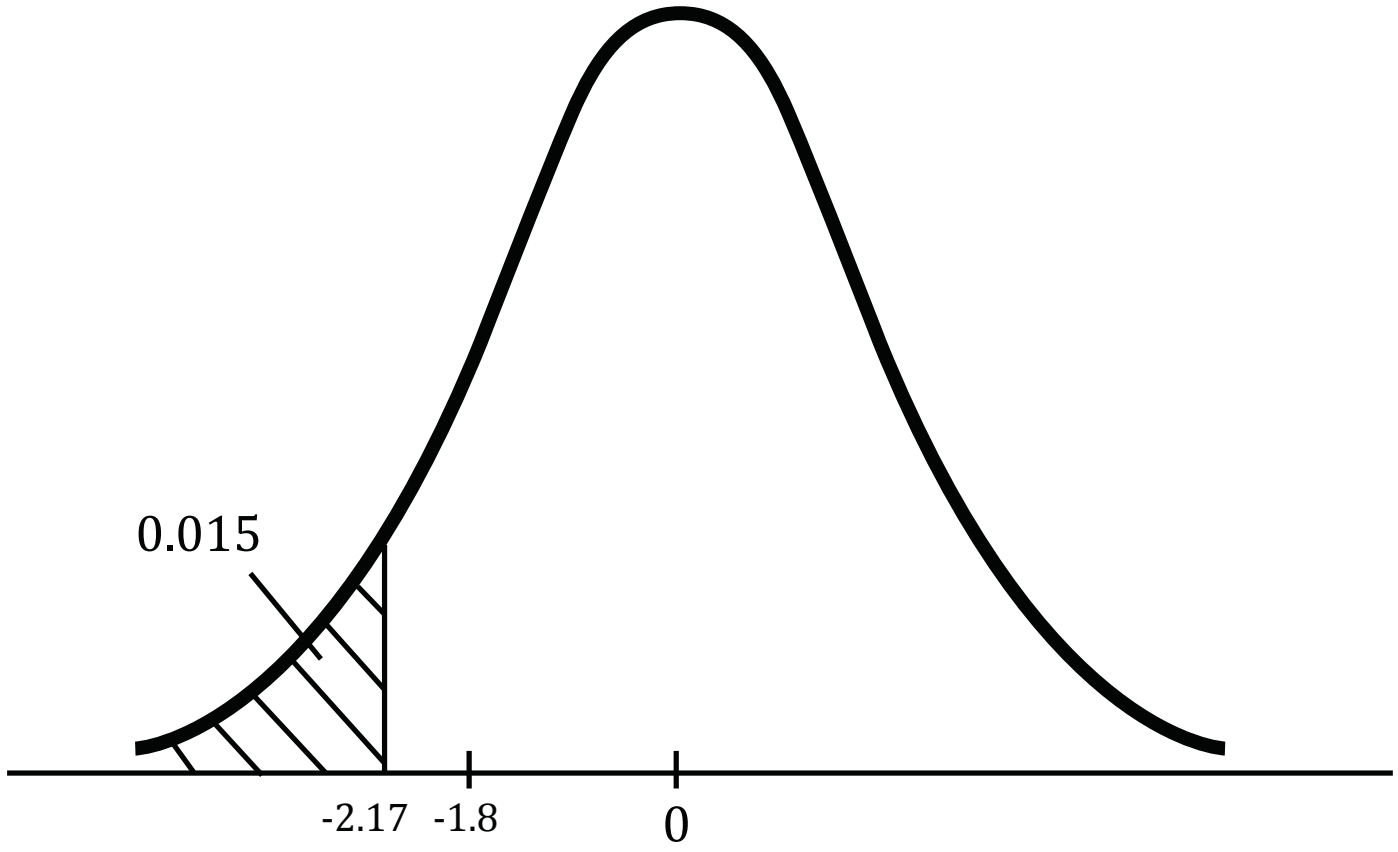
2. The test statistic is $Z = \dfrac{\overline{X} - \mu_0}{\sigma/\sqrt{n}} = \dfrac{\overline{X} - 5}{0.5/\sqrt{36}}$.

   The value of the test statistic is $z = \dfrac{4.85 - 5}{0.5/\sqrt{36}} = -1.8$.

3. **The critical value c is calculated from**

$$P(Z < c) = \alpha = 0.015.$$

From Table A1, we get $c = -2.17$.



4. **Since the test statistic does NOT fall within the critical region, there is no significant evidence against H_0 at 1.5% level. Hence, we will accept H_0 and reject H_1.**

   We conclude that the sample evidence is NOT sufficient to indicate that the new procedure has reduced the mean checking time to below 5 min.

Example 10.2. A random sample of 100 recorded car speeds on a certain road showed an average speed of 71.8 km/h. Assuming a population standard deviation of 8.9 km/h, does this seem to indicate, at the 5% significance level, that the average car speed on that road is greater than 70 km/h?

Soln. Let $X =$ car speed (in km/h). X has a standard deviation of $\sigma = 8.9$ km/h. $\alpha = 0.05$.

Since $n = 100 > 30$, by CLT $\overline{X}$ is approximately normal.

1. $H_0: \ \mu = 70, \qquad H_1: \ \mu > 70$ **(upper-tail test)**

2. **The test statistic is**

$$Z = \frac{\overline{X} - \mu_0}{\sigma/\sqrt{n}} = \frac{\overline{X} - 70}{8.9/\sqrt{100}}.$$

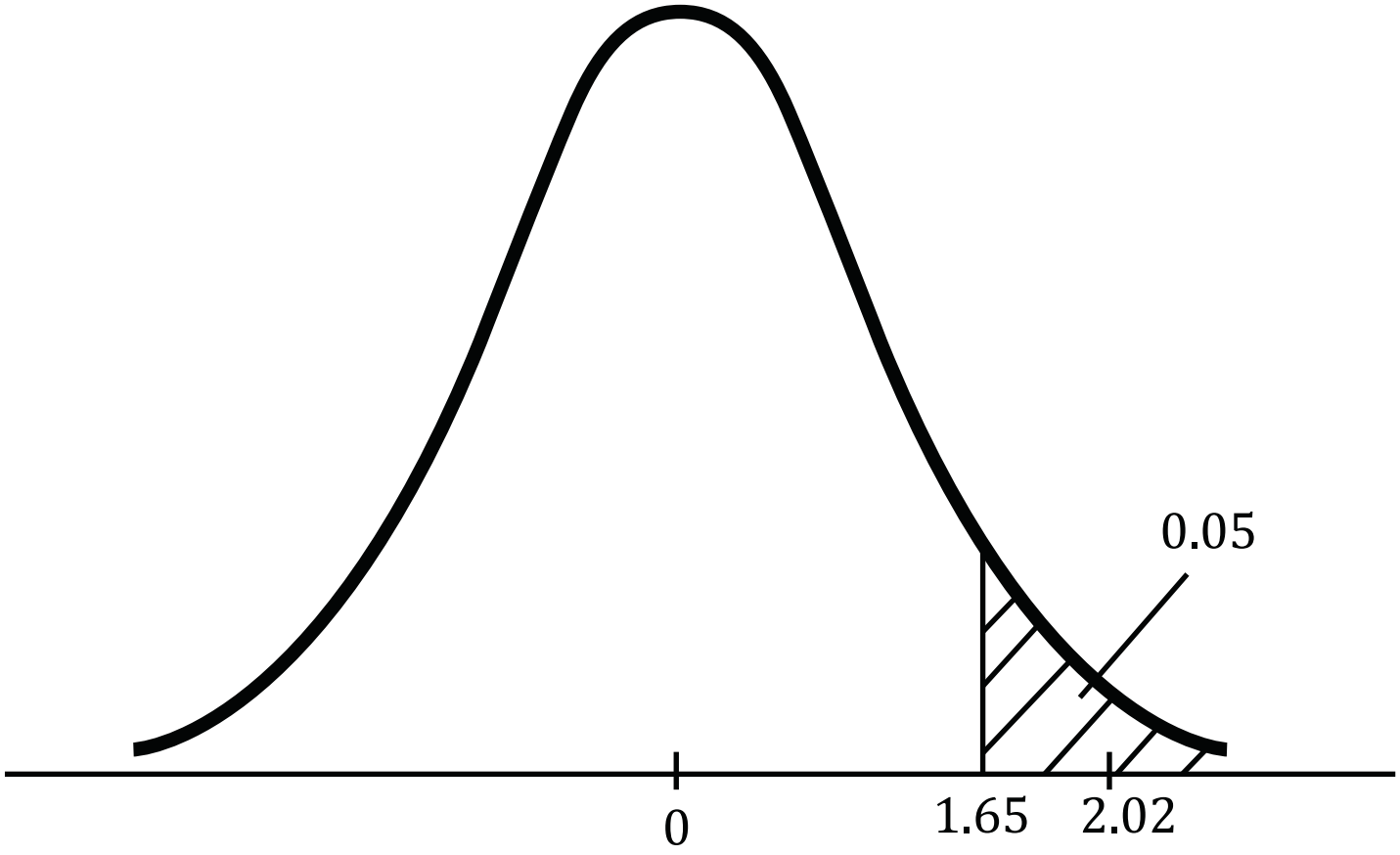
The value of the test statistic is

$$z = \frac{71.8 - 70}{8.9/\sqrt{100}} = 2.02.$$

3. The critical value c is calculated from

$$P(Z > c) = \alpha = 0.05.$$

From Table A1, we get $c = 1.65$.



4. **Since the test statistic falls within the critical region, there is significant evidence against H_0 at 5% level. Hence, we will reject H_0 and accept H_1.**

   We conclude that the sample evidence is sufficient to indicate that the average car speed on that road is greater than 70 km/h.

Example 10.3. Super big fish balls produced in a factory are supposed to average 4.445 cm in diameter. It is known that $\sigma = 0.127$ cm. A random sample of 900 super big fish balls from the production of a single machine gives $\overline{x} = 4.455$ cm. Is the machine within acceptable adjustment? Use $\alpha = 0.05$.

Soln. Let $X =$ diameter of super big fish ball produced by this machine (in cm). X has a standard deviation of $\sigma = 0.127$ cm. $\alpha = 0.05$.

Since $n = 900 > 30$, by CLT $\overline{X}$ is approximately normal. Here $\mu =$ mean diameter of super big fish ball produced by

this machine (in cm).

1. $H_0: \ \mu = 4.445, \qquad H_1: \ \mu \neq 4.445$ (two-tail test)

2. The test statistic is

$$Z = \frac{\overline{X} - \mu_0}{\sigma/\sqrt{n}} = \frac{\overline{X} - 4.445}{0.127/\sqrt{900}}.$$

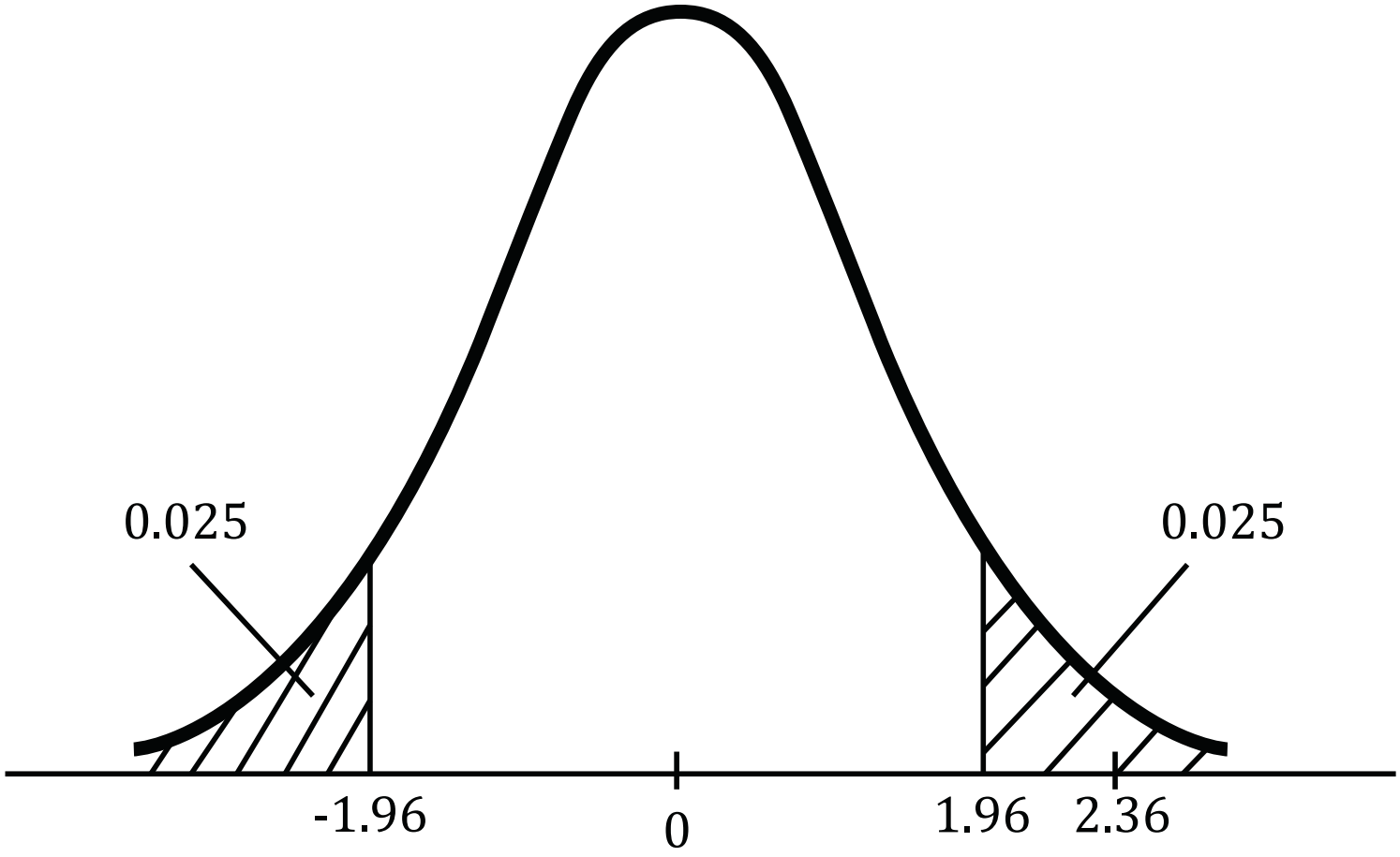
The value of the test statistic is

$$z = \frac{4.455 - 4.445}{0.127/\sqrt{900}} = 2.36.$$

3. The critical values $c_1,\ c_2$ are calculated from

$$P(Z < c_1) = P(Z > c_2) = \frac{\alpha}{2} = 0.025.$$

From Table A1, we get $c_1 = -1.96,\ c_2 = 1.96$.



4. **Since the test statistic falls within the critical region, there is significant evidence against H_0 at 5% level. Hence, we will reject H_0 and accept H_1.**

   We conclude that the sample evidence is sufficient to indicate that the average diameter of super big fish balls produced by this machine is not 4.445 cm. The machine needs to be adjusted.

Example *1. The national average weight of a loaf of wholemeal bread in Singapore is 400 g. A random sample of 20 loafs of Sunshine wholemeal bread gives an average weight of 380 g. The standard deviation of the weights of Sunshine wholemeal loafs is known to be 50 g. Test the hypothesis that the average weight of Sunshine wholemeal loaf is below the national average, using a 5% significance level.

Soln. Let $X =$ weight of a Sunshine wholemeal loaf (in g). X has a standard deviation of $\sigma = 50$. $\alpha = 0.05$.

Since $n = 20 < 30$, we assume that X is approximately

EXTRA

normal so that $\overline{X}$ is also approximately normal.

1. $H_0 : \mu = 400$ $\qquad$ $H_1 : \mu < 400$

2. The test statistic is

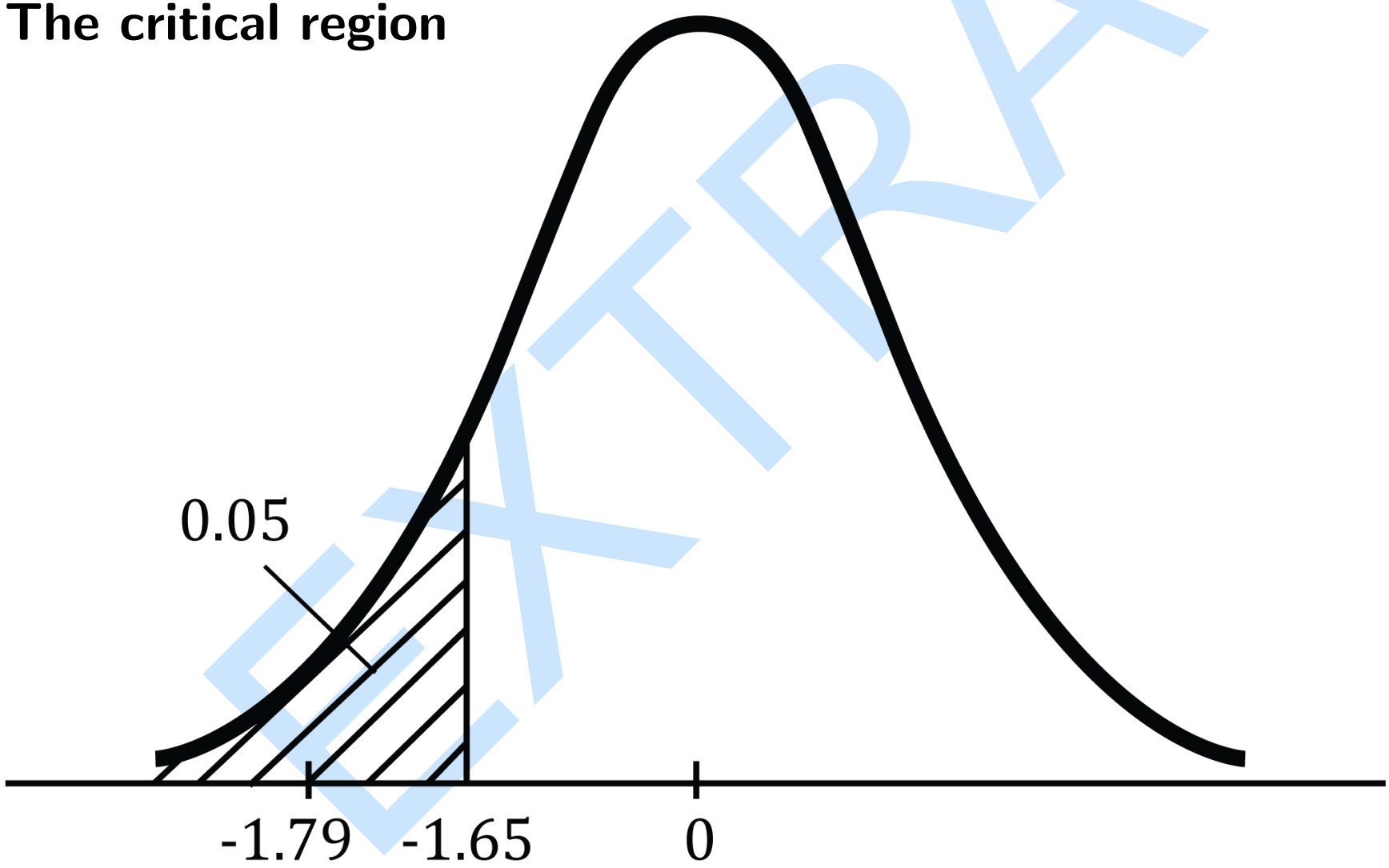
$$Z = \frac{\overline{X} - \mu_0}{\sigma/\sqrt{n}}$$

The value of the test statistic is

$$z = \frac{380 - 400}{50/\sqrt{20}} = -1.79.$$

EXTRA

3. The critical region



EXTRA

4. **Since the test statistic falls within the critical region, we will reject H_0 and accept H_1.**

   We conclude that the sample evidence is sufficient to indicate that the average weight of Sunshine wholemeal loaf is below the national average.

EXTRA

Summary

	Hypothesis testing for μ $\quad H_0 : \mu = \mu_0$
σ known	• require $\overline{X}$ approximately normal • test statistic $Z = \dfrac{\overline{X} - \mu_0}{\sigma/\sqrt{n}}$

10.2. When σ is Unknown

Let X be a r.v. with unknown mean μ and unknown standard deviation σ. We use the sample standard deviation

$$S = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(X_i - \overline{X})^2}$$

in place of σ.

We require

$\overline{X}$ to be approximately normally distributed.

The appropriate test statistic is then

$$t = \frac{\overline{X} - \mu_0}{S/\sqrt{n}}$$

which, assuming H_0 is true, is (approximately) the t random variable with $(n-1)$ d.f.

Recall the steps of hypothesis testing:

(0) Make sure that the requirement is satisfied.

"$\overline{X}$ **is approximately normal**"

(I) Formulate H_0 and H_1

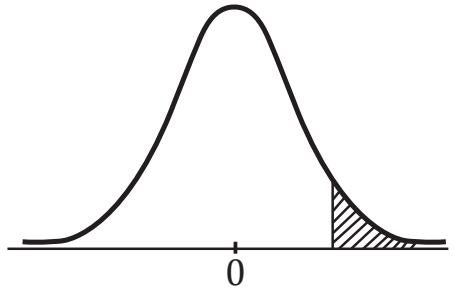
$$H_0 : \mu = \mu_0 \qquad H_1 : \mu \qquad \mu_0$$

(II) Calculate test statistic

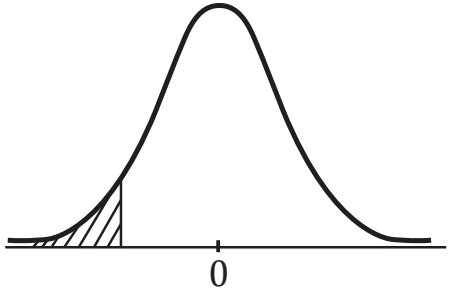
$$t = \frac{\overline{X} - \mu_0}{S/\sqrt{n}}, \qquad d.f. = n - 1$$

(III) Draw rejection region (critical region)

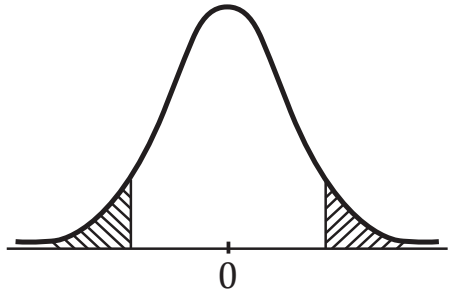
Upper-tail test $H_1 : \mu > \mu_0$



Lower-tail test $H_1 : \mu < \mu_0$



Two-tail test $H_1 : \mu \neq \mu_0$



(IV) Conclusion: reject ? accept ? Write in words about H_1

	Hypothesis testing for μ $\quad H_0 : \mu = \mu_0$
σ unknown	• require $\overline{X}$ approximately normal • test statistic $t = \dfrac{\overline{X} - \mu_0}{S/\sqrt{n}}, \quad d.f. = n - 1$

Example 10.4. A major TV manufacturer wants to test a new model of TV to determine whether it meets the energy saving standard. The mean \$bill μ of a new model TV must be less than 20 cents per hour of usage. 10 TVs are manufactured for testing purpose and the mean and standard deviation of the \$bill for this sample of TVs are determined to be $\overline{x} = 17.1$ cents per hour of usage, $s = 3.0$ cents per hour of usage. Do the data supply sufficient evidence at the 1% level of significance, to allow the manufacturer to conclude that this new model of TV meets the new energy saving standard?

Soln. Let $X =$ \$bill for a new model TV (in cents per hour of usage). $\alpha = 0.01$.

Since $n = 10 < 30$, we assume X is normally distributed so that $\overline{X}$ is normally distributed.

1. The manufacturer wants to establish the alternate hypothesis that the mean \$bill, μ, for all new model TVs is less than 20 cents per hour of usage.

   $H_0 : \ \mu = 20, \qquad H_1 : \ \mu < 20 \qquad$ (lower-tail test)

2. The test statistic is

$$t = \frac{\overline{X} - \mu_0}{S/\sqrt{n}} = \frac{\overline{X} - 20}{3/\sqrt{10}}, \quad d.f. = n - 1 = 9.$$

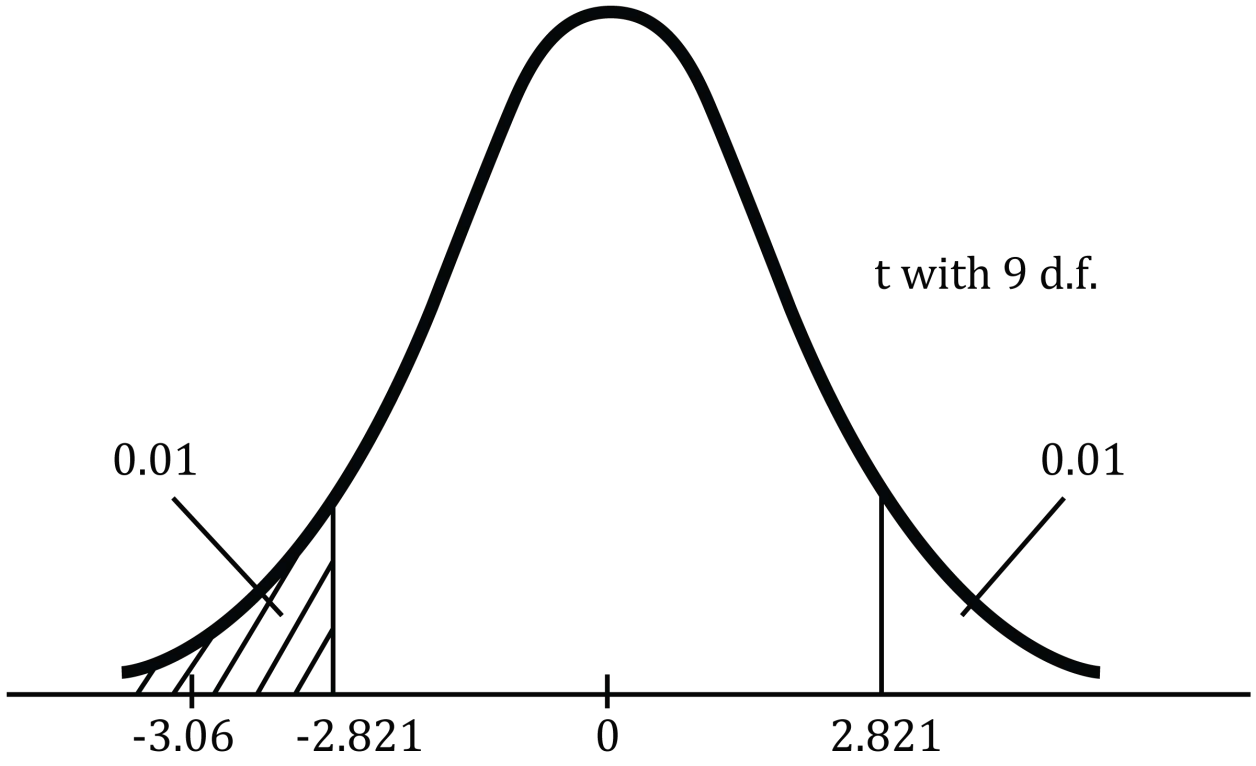
The value of the test statistic is

$$t = \frac{17.1 - 20}{3/\sqrt{10}} = -3.06.$$

3. The critical value c is calculated from

$$P(t < c) = \alpha = 0.01.$$

Since t distribution is symmetrical, we get from Table A3 $c = -t_{0.01} = -2.821$.



4. **Since the test statistic falls within the critical region, there is significant evidence against H_0 at 1% level. Hence, we will reject H_0 and accept H_1.**

   We conclude that the sample evidence is sufficient to indicate that the mean \$bill of the new model TV meets the energy saving standard.

Example 10.5. A new method has been developed to produce synthetic diamonds. 6 diamonds have been produced by the new method, and the mean and standard deviation for the 6 diamonds are

$$\overline{x} = 0.53 \text{ carat}, \qquad s = 0.0559 \text{ carat}.$$

A diamond can be classified as a 'solitaire' if it weighs more than 0.5 carat. Based on the sample of 6 diamonds, can we say that the new method produces solitaires on the average? Use $\alpha = 0.05$ significance level.

Soln. Let X = weight of diamond produced by the new method (in carat). $\alpha = 0.05$.

Since $n = 6 < 30$, we assume X is normally distributed so that $\overline{X}$ is normally distributed.

1. We want to establish the alternate hypothesis that the average weight of diamond, μ, produced by this process is more than 0.5 carat (so that it is a 'solitaire'). Thus,

   $H_0 : \ \mu = 0.5, \qquad H_1 : \ \mu > 0.5$ (upper-tail test)

2. The test statistic is

$$t = \frac{\overline{X} - \mu_0}{S/\sqrt{n}} = \frac{\overline{X} - 0.5}{0.0559/\sqrt{6}}, \quad d.f. = n - 1 = 5.$$

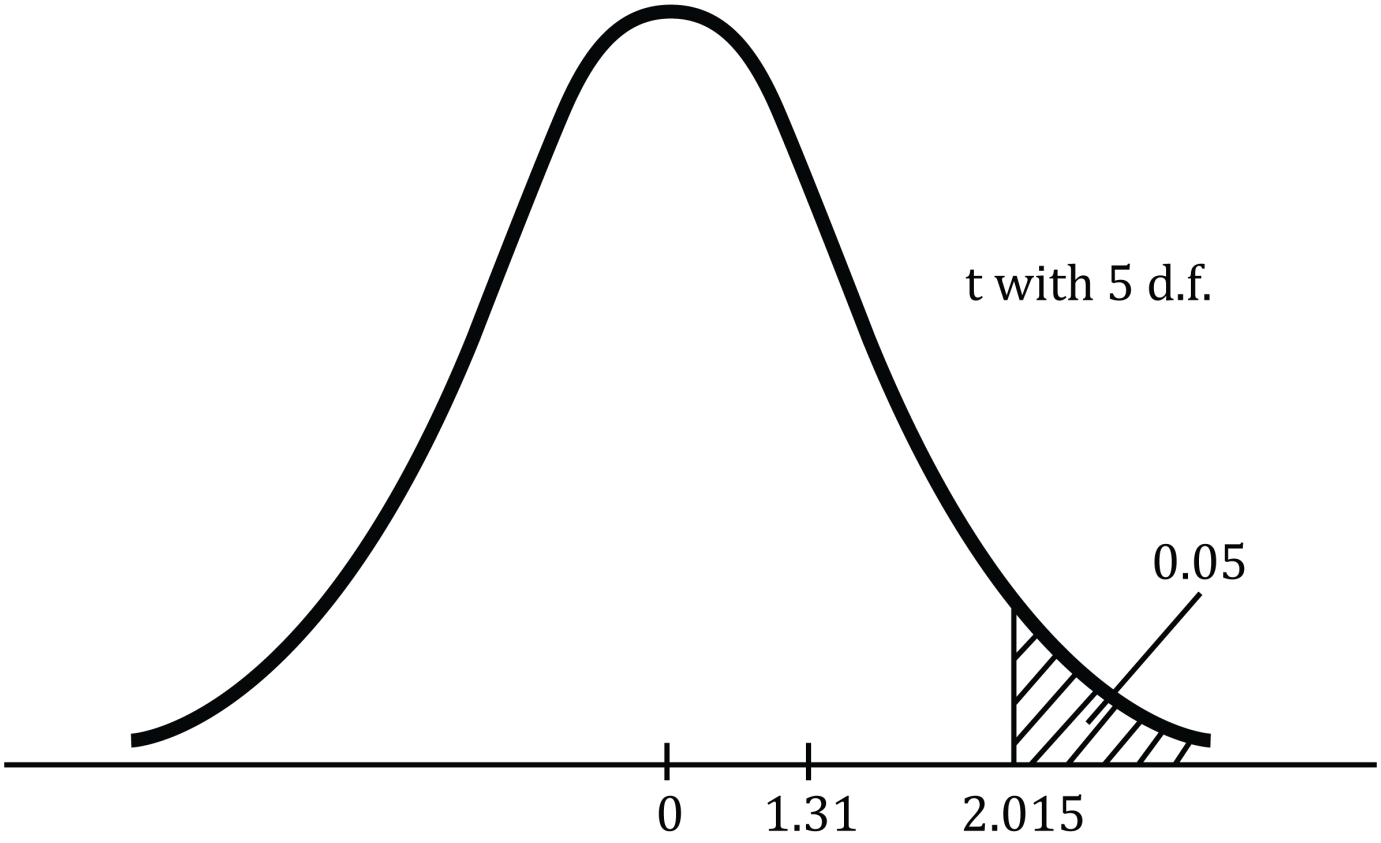
The value of the test statistic is

$$t = \frac{0.53 - 0.5}{0.0559/\sqrt{6}} = 1.31.$$

3. The critical value c is calculated from

$$P(t > c) = \alpha = 0.05.$$

From Table A3, we find $c = t_{0.05} = 2.015$.



4. **Since the test statistic does NOT fall within the critical region, there is no significant evidence against H_0 at 5% level. Hence, we will accept H_0 and reject H_1.**

   We conclude that the sample evidence is NOT sufficient to indicate that the new method can produce solitaires on the average.

Example 10.6. A manufacturer of laser printers monitors the retail price of the printers. For one type of printer the list price is \$750, and the manufacturer wants to know whether the current mean retail price differs from the list price. 17 retailers are sampled and the 17 retail prices give a mean and standard deviation of $\overline{x} = \$732$, $s = \$38$. Does this sample provide sufficient evidence to conclude that the mean retail price differs from the list price of \$750? Use $\alpha = 0.05$ significance level.

Soln. Let $X =$ retail price of printer (in \$). $\alpha = 0.05$.

Since $n = 17 < 30$, we assume X is normally distributed so that $\overline{X}$ is normally distributed.

1. $H_0: \ \mu = 750, \qquad H_1: \ \mu \neq 750$ (two-tail test)

2. The test statistic is

$$t = \frac{\overline{X} - \mu_0}{S/\sqrt{n}} = \frac{\overline{X} - 750}{38/\sqrt{17}}, \quad d.f. = n - 1 = 16.$$

The value of the test statistic is

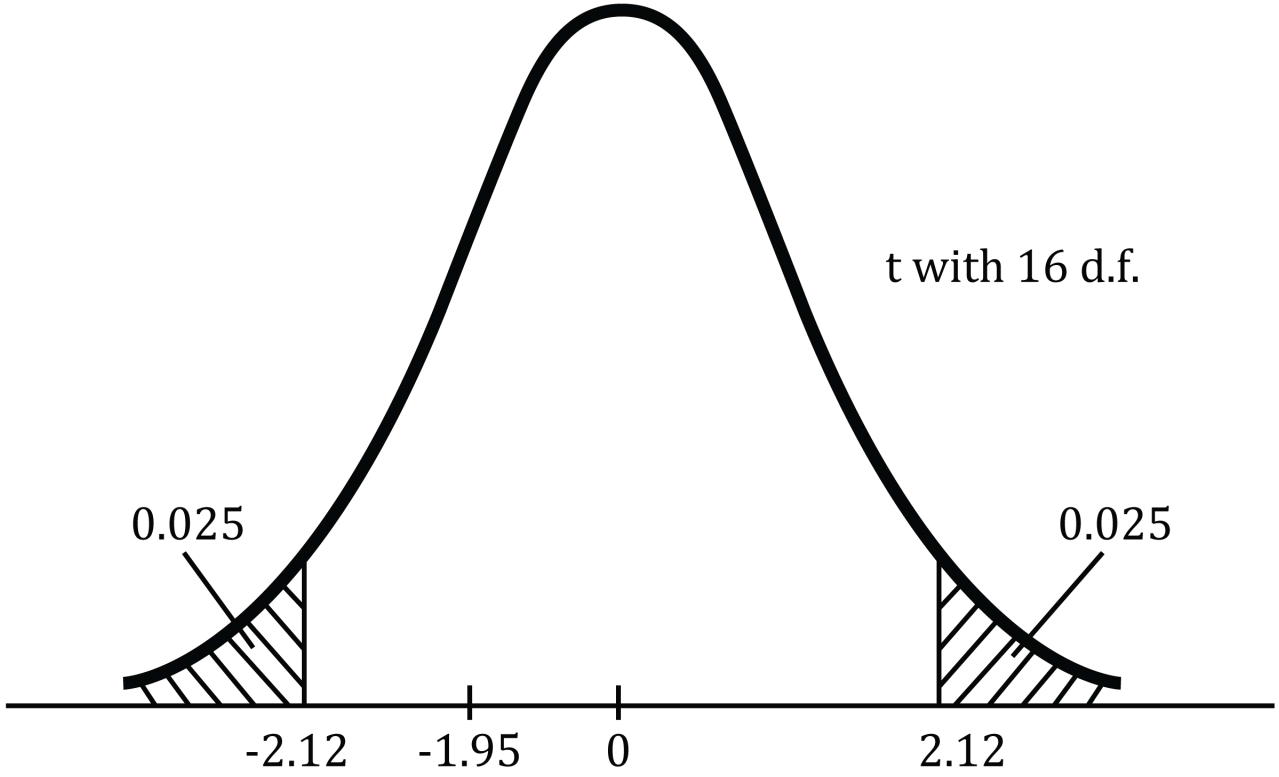
$$t = \frac{732 - 750}{38/\sqrt{17}} = -1.95.$$

3. The critical value $c_1,\ c_2$ are calculated from

$$P(t < c_1) = P(t > c_2) = \frac{\alpha}{2} = 0.025.$$

From Table A3, we find

$$c_2 = t_{0.025} = 2.12, \qquad c_1 = -t_{0.025} = -2.12.$$



4. **Since the test statistic does NOT fall within the critical region, there is no significant evidence against H_0 at 5% level. Hence, we will accept H_0 and reject H_1.**

   Based on the sample evidence, the manufacturer CANNOT conclude that the mean retail price differs from the $750 list price.

Example *2. The average speed of cars travelling on Malaysian express ways is estimated to be 110 km/h. A random sample of 41 cars travelling on a Singapore express way, PIE, gives an average speed of 100 km/h and a standard deviation of 15 km/h. Can we say, at 2.5% significance level, that cars drive slower on PIE than on Malaysian express ways?

Soln. Let $X =$ speed of a car on PIE (in km/h). $\alpha = 0.025$.

Since $n = 41 > 30$, by CLT we know that $\overline{X}$ is approximately normal.

EXTRA

1. $H_0 : \mu = 110$ $\quad\quad$ $H_1 : \mu < 110$

2. The test statistic is

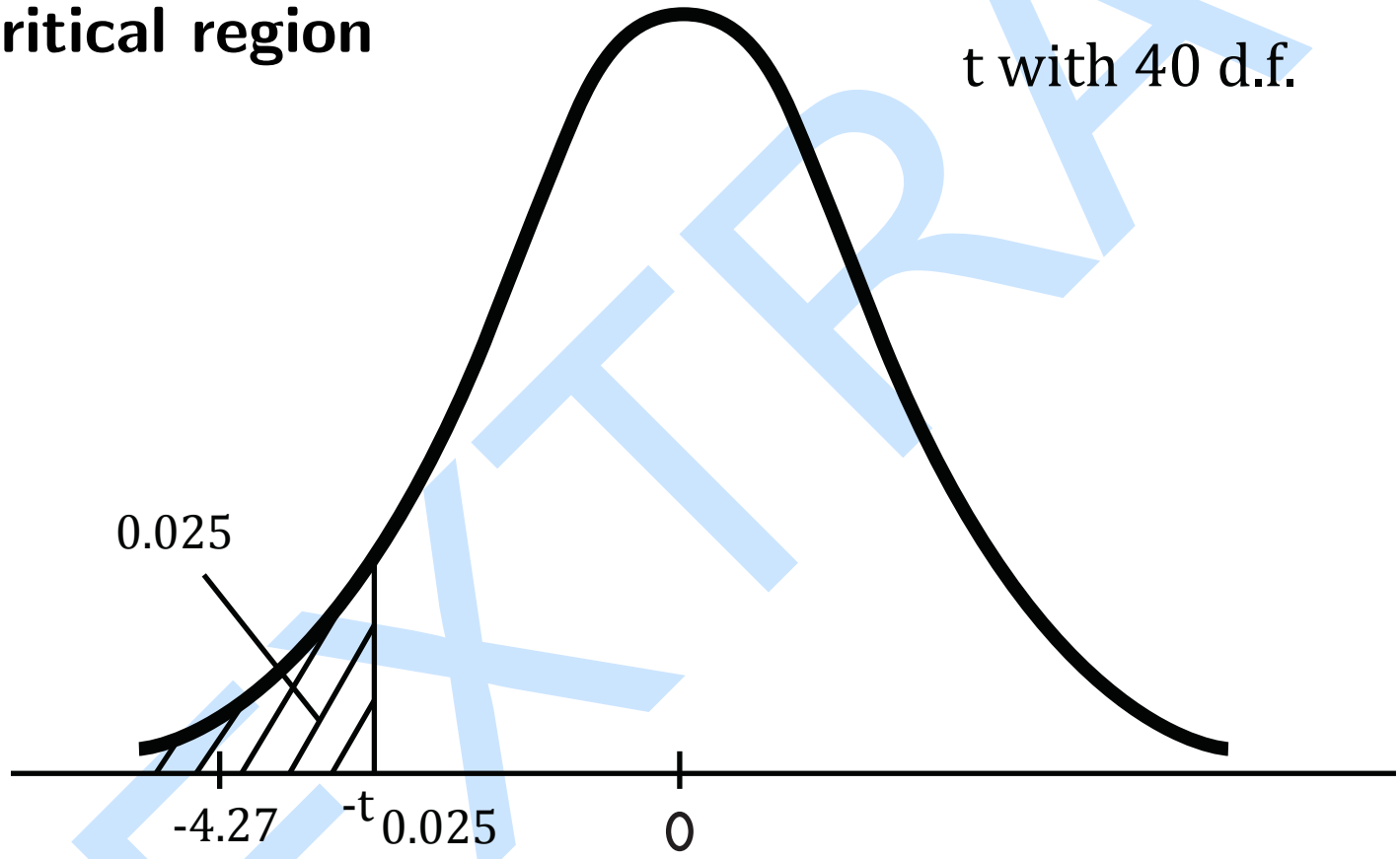
$$t = \frac{\overline{X} - \mu_0}{S/\sqrt{n}}, \quad d.f. = n - 1$$

The value of the test statistic is

$$t = \frac{100 - 110}{15/\sqrt{41}} = -4.27.$$

EXTRA

3. The critical region



$$-t_{0.025} = -2.021$$

EXTRA

4. **Since the test statistic falls within the critical region, we will reject H_0 and accept H_1.**

   We conclude that the sample evidence is sufficient to indicate that the average speed of cars on PIE is less than that on Malaysian express ways.

EXTRA

Summary

	Hypothesis testing for μ $\quad H_0 : \mu = \mu_0$
σ unknown	• require $\overline{X}$ approximately normal • test statistic $t = \dfrac{\overline{X} - \mu_0}{S/\sqrt{n}}, \quad d.f. = n - 1$

	Hypothesis testing for μ $\quad H_0 : \mu = \mu_0$
σ known	• require $\overline{X}$ approximately normal • test statistic $Z = \dfrac{\overline{X} - \mu_0}{\sigma/\sqrt{n}}$
σ unknown	• require $\overline{X}$ approximately normal • test statistic $t = \dfrac{\overline{X} - \mu_0}{S/\sqrt{n}}, \quad d.f. = n - 1$

11. Tests Concerning the Population Variance σ^2

Let X be a r.v. with unknown mean μ and unknown standard deviation σ. We require

X to be approximately normally distributed.

The null hypothesis is

$$H_0: \quad \sigma^2 = \sigma_0^2.$$

Let α be the specified level of significance for the test. Complete the following steps.

1. **State null and alternative hypotheses:** $H_0: \sigma^2 = \sigma_0^2$, $H_1: \sigma^2 < \sigma_0^2$ or $H_1: \sigma^2 > \sigma_0^2$ or $H_1: \sigma^2 \neq \sigma_0^2$.

2. **Compute test statistic:** The test statistic is

$$\chi^2 = \frac{(n-1)S^2}{\sigma_0^2}$$

which, assuming H_0 is true, is (approximately) a chi-square r.v. with $(n-1)$ d.f. Compute value of the test statistic

$$\hat{\chi}^2 = \frac{(n-1)s^2}{\sigma_0^2}$$

where s^2 is the sample variance

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \overline{x})^2.$$

3. Determine critical region: Using a table for the chi-square r.v. with $(n-1)$ d.f. (Table A2), find the critical value and critical region corresponding to H_1. There are 3 different situations.

- **Upper-tail test** $H_0: \sigma^2 \leq \sigma_0^2$, $H_1: \sigma^2 > \sigma_0^2$

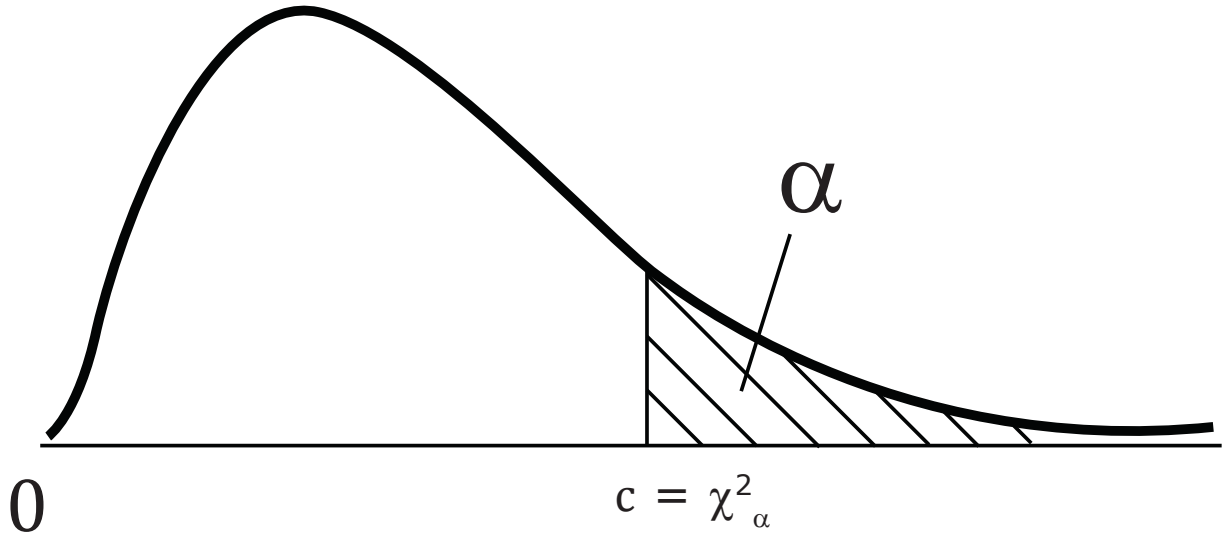
  The rejection region is given by the set of outcomes

  $$\{\chi^2 > c\}$$

  where the value c, called the critical value, is defined by

  $$P(\chi^2 > c) = \alpha.$$

  Clearly, $c = \chi^2_\alpha$.



- **Lower-tail test** $H_0: \sigma^2 \geq \sigma_0^2$, $H_1: \sigma^2 < \sigma_0^2$

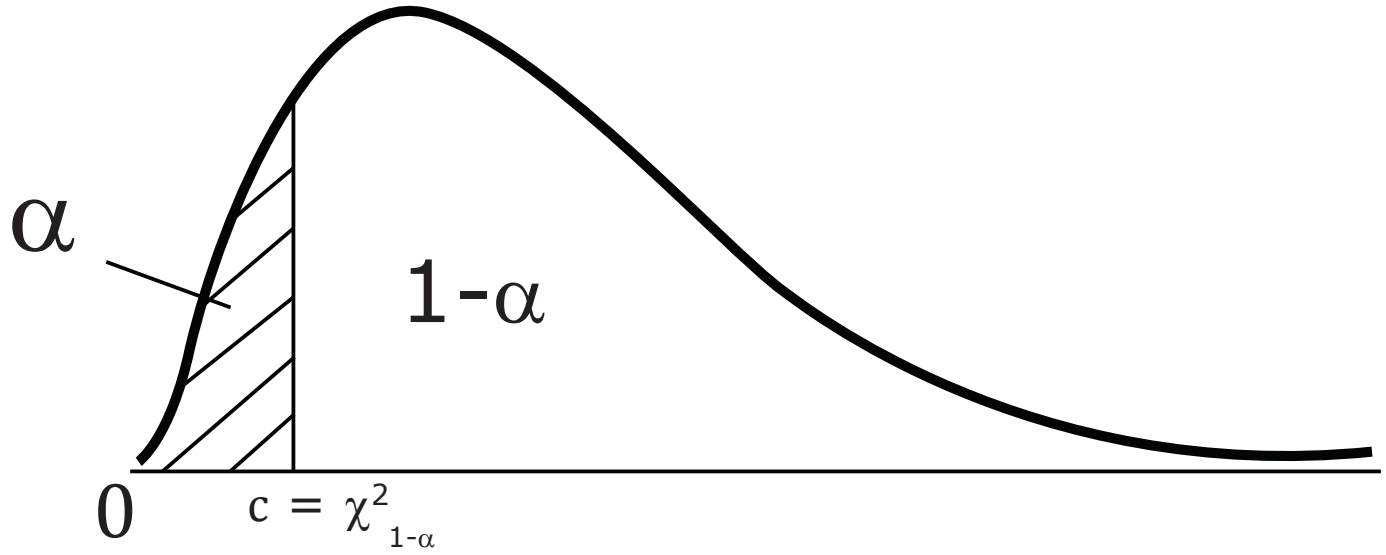
  The rejection region is given by the set of outcomes

  $$\{\chi^2 < c\}$$

  where the critical value c is defined by

  $$P(\chi^2 < c) = \alpha.$$

  Clearly, $c = \chi^2_{1-\alpha}$.



- **Two-tail test** $H_0: \sigma^2 = \sigma_0^2, \qquad H_1: \sigma^2 \neq \sigma_0^2$

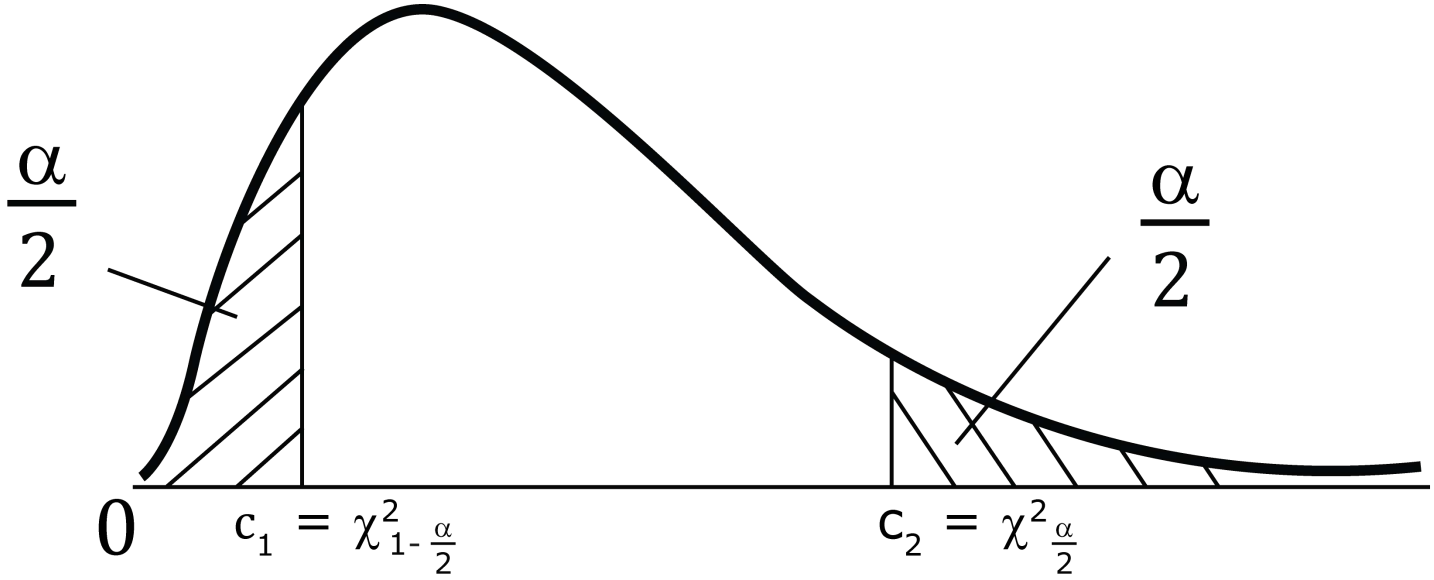
  The rejection region is given by the set of outcomes

$$\{\chi^2 < c_1\} \cup \{\chi^2 > c_2\}$$

  where the critical values $c_1,\ c_2$ are defined by

$$P(\chi^2 < c_1) = P(\chi^2 > c_2) = \frac{\alpha}{2}.$$

  Note that chi-square distribution is non-symmetrical, hence $c_1 \neq -c_2$. Clearly, $c_2 = \chi^2_{\alpha/2}$, $c_1 = \chi^2_{1-\alpha/2}$.



4. **Draw conclusion:** If the sample value, $\hat{\chi}^2$, of the test statistic lies in the critical region, then H_0 is rejected at the level of significance α. If $\hat{\chi}^2$ does NOT lie in the critical region, then H_0 is accepted at the level of significance α.

We summarize the steps of hypothesis testing:

(0) Make sure that the requirement is satisfied.

"X is approximately normal"

(I) Formulate H_0 and H_1

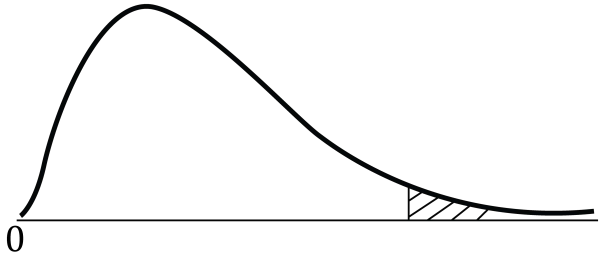
$$H_0 : \sigma^2 = \sigma_0^2 \qquad H_1 : \sigma^2 \qquad \sigma_0^2$$

(II) Calculate test statistic

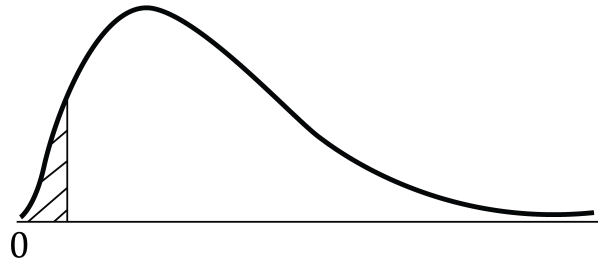
$$\chi^2 = \frac{(n-1)S^2}{\sigma_0^2}$$

(III) Draw rejection region (critical region)

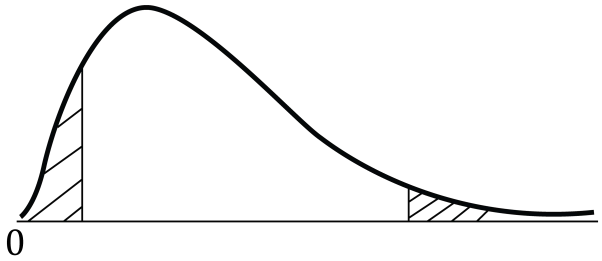
Upper-tail test $H_1 : \sigma^2 > \sigma_0^2$



Lower-tail test $H_1 : \sigma^2 < \sigma_0^2$



Two-tail test $H_1 : \sigma^2 \neq \sigma_0^2$



(IV) Conclusion: reject ? accept ? Write in words about H_1

Hypothesis testing for σ^2 $\quad H_0 : \sigma^2 = \sigma_0^2$

- require X approximately normal
- test statistic $\chi^2 = \dfrac{(n-1)S^2}{\sigma_0^2}$

Example 11.1. The amount of gold used to plate a necklace (costume jewelery) is normally distributed with standard deviation of $\sigma = 1.2$ g. A new plating procedure is designed to decrease the variability of the amount of gold used. A sample of 101 necklaces gold-plated by the new procedure has a standard deviation of 0.98 g. Can we conclude that the new plating procedure decrease the variability of the amount of gold used, at 2.5% level of significance?

Soln. Let $X =$ the amount of gold used to plate a necklace (in g). X is normal. $\alpha = 0.025$.

1. We want to establish the alternate hypothesis that the standard deviation of the amount of gold used in the new procedure is less than 1.2 g. Thus,

$H_0: \ \sigma^2 = 1.44, \qquad H_1: \ \sigma^2 < 1.44$ **(lower-tail test)**

2. The test statistic is

$$\chi^2 = \frac{(n-1)S^2}{\sigma_0^2} = \frac{100S^2}{1.44}, \quad d.f. = n - 1 = 100.$$

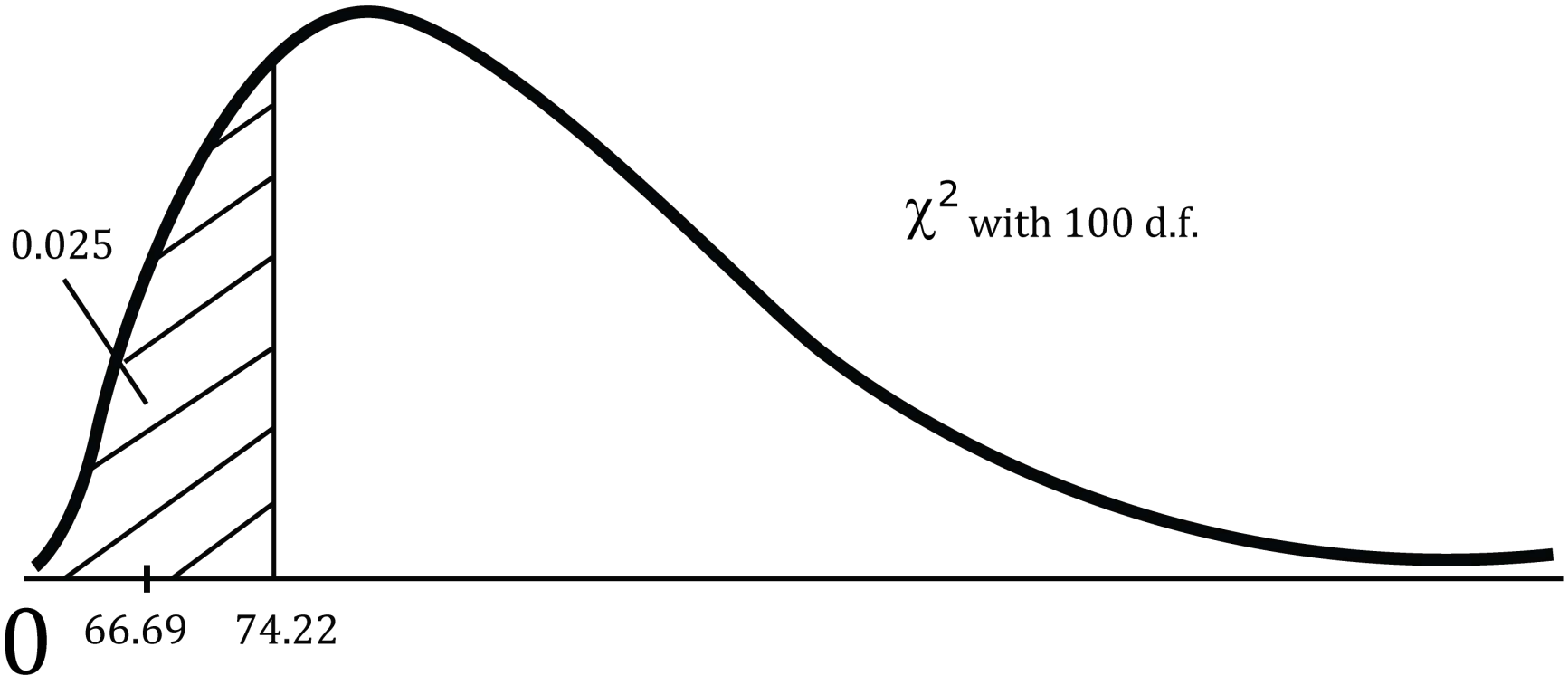
The value of the test statistic is

$$\hat{\chi}^2 = \frac{100(0.98)^2}{1.44} = 66.69.$$

3. The critical value c is calculated from

$$P(\chi^2 < c) = \alpha = 0.025.$$

From Table A2, we get $c = \chi^2_{0.975} = 74.22$.



4. **Since the test statistic falls within the critical region, there is significant evidence against H_0 at 2.5% level. Hence, we will reject H_0 and accept H_1.**

   We conclude that the new plating procedure does decrease the variability of the amount of gold used.

Example 11.2. Over the years the weights of apple pies sold in a fast food restaurant have been normally distributed with mean 75 g and standard deviation of 8 g. Recently, the apple pies seem to get smaller and their weights seem to show more variation. A sample of 41 apple pies has mean $\overline{x} = 73$ g and standard deviation of $s = 9.6$ g. Assuming that the weights of apple pies are still normally distributed, test the hypothesis that the variation of the weights has increased, using a 5% significance level.

Soln. Let $X =$ the weight of an apple pie (in g). X is normal.

$\alpha = 0.05.$

1. $H_0: \ \sigma^2 = 64, \qquad H_1: \ \sigma^2 > 64$ **(upper-tail test)**

2. **The test statistic is**

$$\chi^2 = \frac{(n-1)S^2}{\sigma_0^2} = \frac{40S^2}{64}, \quad d.f. = n - 1 = 40.$$

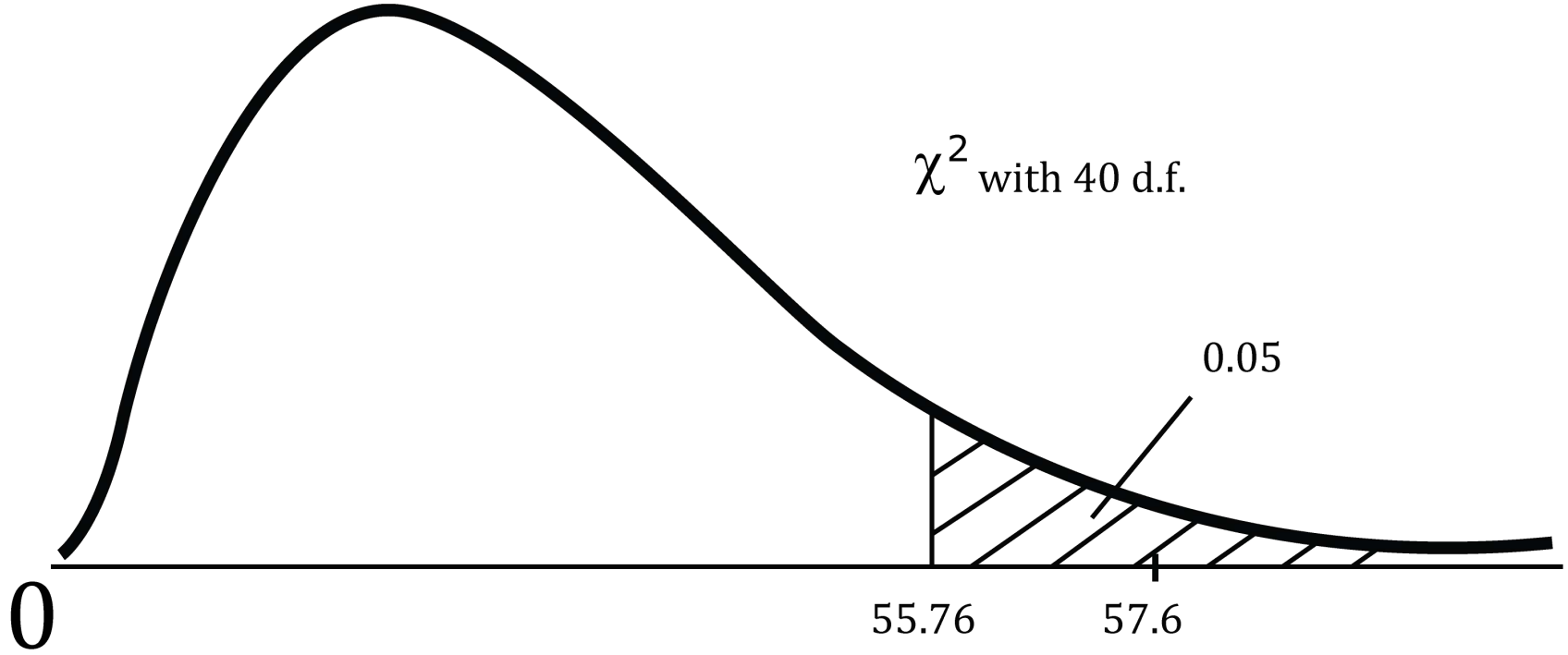
The value of the test statistic is

$$\hat{\chi}^2 = \frac{40(9.6)^2}{64} = 57.6.$$

3. The critical value c is calculated from

$$P(\chi^2 > c) = \alpha = 0.05.$$

From Table A2, we get $c = \chi^2_{0.05} = 55.76$.



4. **Since the test statistic falls within the critical region, there is significant evidence against H_0 at 5% level. Hence, we will reject H_0 and accept H_1.**

   We conclude that the the variation of the weights has increased.

Example 11.3. The number of glasses of water consumed by a female daily is a normal r.v. with mean $\mu = 7.5$ and variance $\sigma^2 = 1.25$. The daily water consumption of males may be different. A sample of 15 males on a day gives an average of $\overline{x} = 6.25$ and variance of $s^2 = 1.5$ (number of glasses of water). Assuming that the number of glasses of water consumed daily is still normally distributed, test the hypothesis that the variance of the number of glasses of water consumed daily by a male is different from that consumed daily by a female, using a 5% significance level.

Soln. Let $X =$ the number of glasses of water consumed daily by a male. X is normal. $\alpha = 0.05$.

1. $H_0: \ \sigma^2 = 1.25, \qquad H_1: \ \sigma^2 \neq 1.25$ (two-tail test)

2. The test statistic is

$$\chi^2 = \frac{(n-1)S^2}{\sigma_0^2} = \frac{14S^2}{1.25}, \quad d.f. = n - 1 = 14.$$

The value of the test statistic is

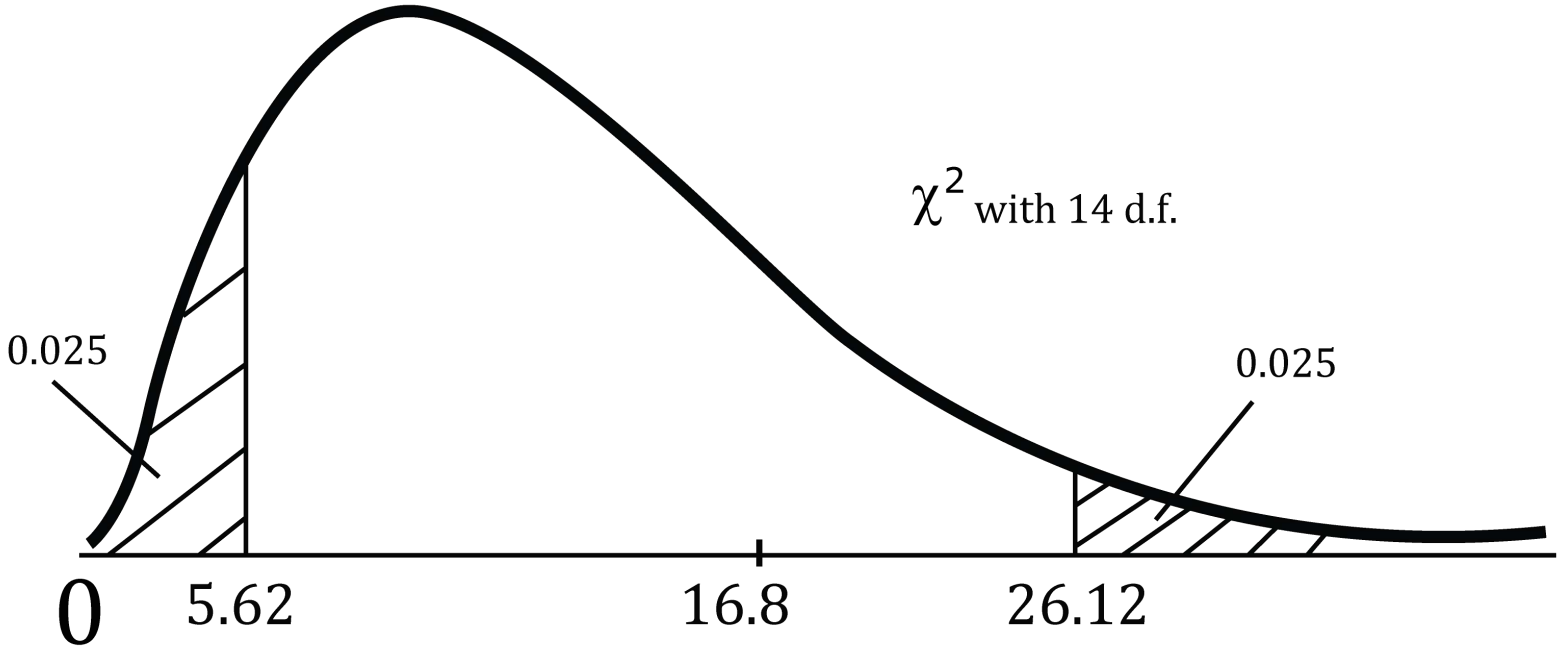
$$\hat{\chi}^2 = \frac{14(1.5)}{1.25} = 16.8.$$

3. The critical values $c_1,\ c_2$ are calculated from

$$P(\chi^2 < c_1) = \frac{\alpha}{2} = 0.025 \quad \textbf{and} \quad P(\chi^2 > c_2) = \frac{\alpha}{2} = 0.025.$$

From Table A2, we get

$$c_1 = \chi^2_{0.975} = 5.62, \qquad c_2 = \chi^2_{0.025} = 26.12.$$



4. Since the test statistic does NOT fall within the critical region, there is no significant evidence against H_0 at 5% level. Hence, we will accept H_0 and reject H_1.

   We conclude that the variance of the number of glasses of water consumed daily by a male is NOT different from that consumed daily by a female.

Example *1. An engineer at a airplane service centre has designed a new procedure that he thinks will reduce the variation in servicing times of Boeing planes. The historical standard deviation of Boeing plane servicing time is known to be 1.40 hours. In an experiment with the new procedure, a random sample of 21 Boeing planes gives an average of 8.35 hours for servicing with a standard deviation of 1.23 hours. Can the engineer claim, at 1% level of significance, that the new procedure has reduced the standard deviation of Boeing plane servicing time?

EXTRA

Soln. Let $X =$ servicing time of a Boeing plane (in hours).

Assume that X is approximately normal. $\alpha = 0.01$.

1. $H_0: \ \sigma^2 = 1.40^2, \qquad H_1: \ \sigma^2 < 1.40^2$ (lower-tail test)

2. The test statistic is

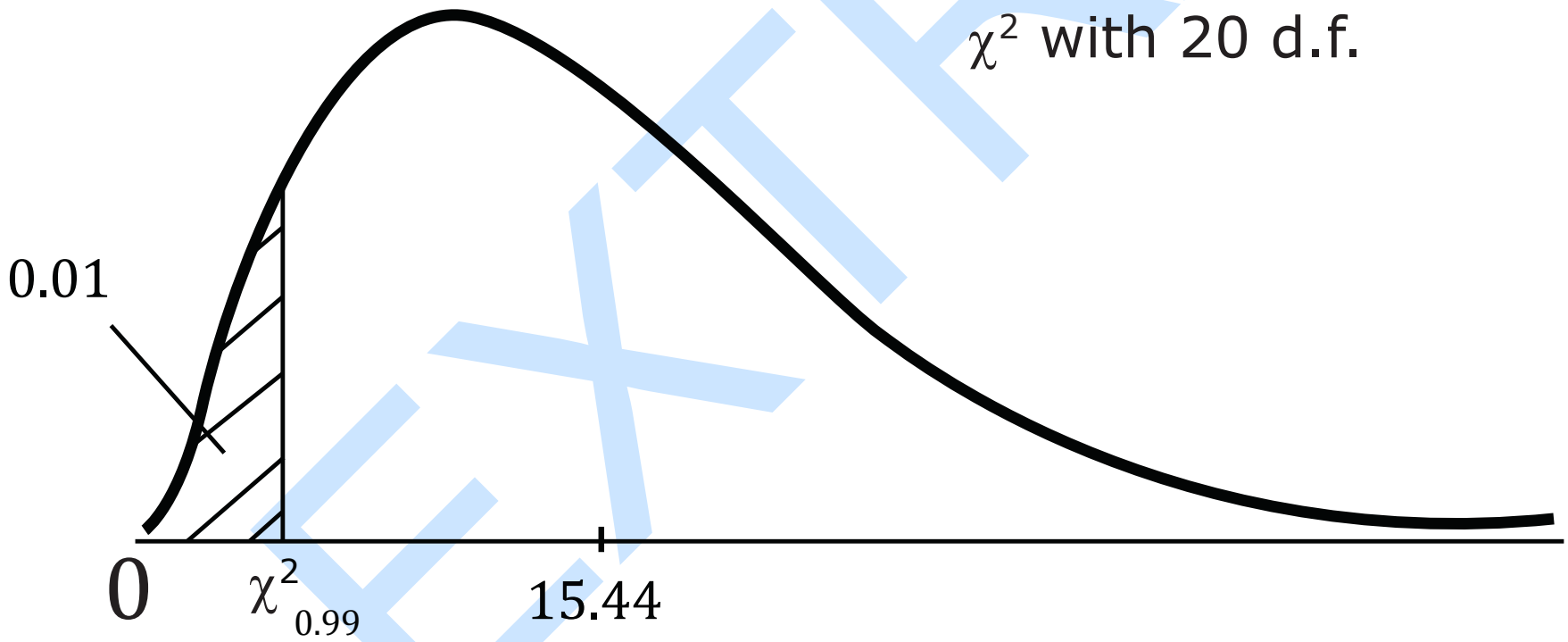
$$\chi^2 = \frac{(n-1)S^2}{\sigma_0^2} = \frac{20S^2}{1.40^2}, \quad d.f. = n - 1 = 20.$$

The value of the test statistic is

$$\hat{\chi}^2 = \frac{20(1.23)^2}{1.40^2} = 15.44.$$

EXTRA

3. The critical value $c = \chi^2_{0.99} = 8.26$



EXTRA

4. **Since the test statistic does NOT fall within the critical region, we will accept H_0 and reject H_1.**

   The engineer cannot claim, at 1% level of significance, that the new procedure has reduced the standard deviation of Boeing plane servicing time.

EXTRA

Summary

Hypothesis testing for σ^2 $\quad H_0 : \sigma^2 = \sigma_0^2$

- require X approximately normal
- test statistic $\chi^2 = \dfrac{(n-1)S^2}{\sigma_0^2}$